INTERMEDIATE COURSE

GROUP – II

REVISION TEST PAPERS

JANUARY, 2026

BOARD OF STUDIES (ACADEMIC)

THE INSTITUTE OF CHARTERED ACCOUNTANTS OF INDIA

(Set up by an Act of Parliament)

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REVISION TEST PAPER, JANUARY, 2026– OBJECTIVE & APPROACH

(Students are advised to go through the following paragraphs carefully to derive maximum benefit out of this RTP)

I. Objective of Revision Test Paper

Revision Test Papers are one among the many educational inputs provided by the Board of Studies (Academic) to its students. Popularly referred to as RTP by the students, it is one of the very old publications of the BOS(A) whose significance and relevance from the examination perspective has stood the test of time.

The primary objectives of the RTP are:

- To help students get an insight of their preparedness for the forthcoming examination;
- To update them on the latest developments relevant for the forthcoming examination in select subjects;
- To enhance the confidence level of the students adequately.

Students must bear in mind that the RTP contains a variety of questions based on different topics of the syllabi and thus a comprehensive study of the entire syllabus is a pre-requisite before answering the questions of the RTP. In other words, in order to derive maximum benefit out of the RTPs, it is advised that before proceeding to solve the questions given in the RTP, students ought to have thoroughly read the Study Materials and Statutory Update, wherever applicable.

The topics on which the questions are set herein have been carefully selected and meticulous attention has been paid in framing different types of questions. Detailed answers are provided to enable the students to do a self-assessment and have a focused approach for effective preparation.

Live Virtual Classes by renowned subject experts conducted free of charge for the students of Foundation, Intermediate and Final levels provide the students much required support in preparing for their exams conveniently at home as these classes can be accessed live or viewed later as recorded lectures through hand-held devices such as smart phones, laptops, I-pads, tablets, etc. anytime anywhere. Further, students are advised to attempt the Multiple-Choice Questions (MCQs) at MCQ Paper Practice Portal which is a holistic platform for self-assessment within the stipulated timeframe.

Students are welcome to send their suggestions for fine tuning the RTP to the Joint Director, Board of Studies (Academic), The Institute of Chartered Accountants of India, A-29, Sector-62, Noida 201309 (Uttar Pradesh). RTP is also available on BOS Knowledge Portal at *https://boslive.icai.org* for downloading.

II. Planning and preparing for examination

Ideally, when the RTP reaches your hand, you must have finished reading the relevant Study Materials of all the subjects available at the BoS Knowledge Portal. Get a good grasp of the concepts/ provisions/ amendments/ cases discussed therein.

After reading the Study Materials alongwith Statutory Update thoroughly, then, proceed to solve the questions given in the RTP on your own. RTP is an effective tool to revise and refresh the concepts and provisions discussed in the Study Material. RTPs are provided to you to help you assess your level of preparation. Hence you must solve the questions given therein on your own and thereafter compare your answers with the answers given therein.

Examination tips

How well a student fares in the examination depends upon the level and depth of his preparation. However, there are certain important points which can help a student better his performance in the examination. These useful tips are given below:

Reach the examination hall well in time.

As soon as you get the question paper, read it carefully and thoroughly. You are given separate 15 minutes for reading the question paper.

Plan your time so that appropriate time is awarded for each question.

First impression is the last impression. The question which you can answer in the best manner should be attempted first.

Always attempt to do all questions. Therefore, it is important that you must finish each question within allocated time. Keep sometime for checking the answers as well.

Read the question carefully more than once before starting the answer to understand very clearly as to what is required.

Answer all parts of a question one after the other; do not answer different parts of the same question at different places.

Write in a neat and legible hand-writing.

Always be concise and write to the point and do not try to fill pages unnecessarily.

There must be logical expression of the answer.

In case a question is not clear, you may state your assumptions and then answer the question.

Check your answers carefully and underline important points before leaving the examination hall.

In case of case scenario based MCQs, read the facts given in the case attentively. Also, read each MCQ based thereon and all the options carefully, before choosing the correct answer.

III. Subject-wise Applicability

PAPER – 4: COST AND MANAGEMENT ACCOUNTING

The Revision Test Paper (RTP) of Cost and Management Accounting comprises of two Divisions of total 19 questions: Division A (Case scenario and caselet based MCQs) which contains 7 case scenarios and caselets and Division B (Descriptive questions) which contains 12 questions for full coverage of the syllabus. Theoretical questions along with computational problems have also been incorporated so that you are able to give emphasis to the theoretical portion of the syllabus as well. Since this paper's inclination is more towards numerical-oriented questions which involve mathematical calculations, therefore, it is very important that you have thoroughly studied the theoretical aspects of the subject and are also clear about the concepts and logic behind the mathematical workings and formulae.

A summary of the questions both theoretical and computational has been given for your reference:

Q. No.	Topic	About the Problem
Division A		
1.	Service Costing	Calculation of total cost and freight.
2.	Cost Sheet	Calculation of various items of cost sheet.
3.	Standard Costing	Calculation of variances.
4.	Cost Sheet	Calculation of selling price per unit.
5.	Standard Costing	Calculation of material variances.
6.	Marginal Costing	Calculation of Minimum price.
7.	Budget and Budgetary Control	Calculation of no. of units purchased
Division B		
8.	Material Cost	Calculation of purchase cost, EOQ and best purchase option.
9.	Employee Cost	Calculation of Earnings.
10.	Overheads	Preparation of cost distribution.
11.	Activity Based Costing	Profitability analysis of customers.
12.	Cost Accounting System	Preparation of costing P&L, control accounts and Reconciliation.
13.	Job and Batch Costing	Compute the cost and price for the new job.
14.	Process Costing	Preparation of process accounts.
15.	Joint product and By product	Calculation of joint cost and profit with further processing decision.
16.	Standard Costing	Calculation of Material variances.

17.	Marginal Costing	Whether to make or buy.
18.	Budget and Budgetary Control	Preparation of material purchase and wage budget.
19(a)	Introduction to cost and management accounting	Types of cost in managerial decision making.
19(b)	Activity Based Costing	Cost drivers for the cost pools.
19(c)	Budget and Budgetary Control	Limitations of Budgetary Control System,
19(d)	Introduction to cost and management accounting	Cost control vs Cost reduction.

PAPER – 5: AUDITING AND ETHICS

The Revisionary Test Paper (RTP) is a tool to refresh your knowledge which you have acquired while doing your conceptual study from Study Material and other modes of knowledge like BoS Live Learning Classes, Webinars, Saransh – Last Mile Referencer, Case Scenarios Booklet, Model Test Papers, Students' Journal, Bare Acts etc.

RTP of Auditing and Ethics for January 2026 comprises twenty questions, including case scenario-based multiple-choice questions, independent multiple choice questions, and descriptive questions. These questions cover the entire syllabus, which is divided into eleven chapters as discussed in the study material.

These 20 questions are taken from different topics like Nature, Objective and Scope of Audit, Audit Strategy, Audit Planning and Audit Program, Risk Assessment and Internal Control, Audit Evidence, Audit of Items of Financial Statements, Audit Documentation, Completion and Review, Audit Report, Special Features of Audit of Different Type of Entities, Audit of Banks and Ethics and Terms of Audit Engagements. The chapter's name is clearly indicated before each question. The questions in the RTP have been arranged in the same sequence as prescribed in the study material to facilitate easy revision by the students. An attempt has been made to cover the syllabus comprehensively.

PAPER – 6: FINANCIAL MANAGEMENT AND STRATEGIC MANAGEMENT

Section – A: Financial Management

The Revision Test Paper (RTP) of Financial Management comprises of two Divisions of total 11 questions: Division A (Case scenario based MCQs) which contains 1 case scenario and 2 caselets and Division B (Descriptive questions) which contains 8 questions for full coverage of the syllabus. Theoretical questions along with computational problems have also been incorporated so that you are able to give emphasis to the theoretical portion of the syllabus as well. Since this paper's inclination is more towards numerical-oriented questions which involve mathematical calculations, therefore, it is very important that you have thoroughly studied the theoretical aspects of the subject and are also clear about the concepts and logic behind the mathematical workings and formulae.

A summary of the questions both theoretical and computational has been given for your reference:

Q. No.	Topic	About the Problem
Division A		
1.	Working Capital	Calculation of working capital.
2.	Investment Decision	Calculation of NPV and Desirability Factor.
3.	Ratio Analysis	DU Pont Analysis.
4.	Leverage	Calculation of Interest coverage ratio.
Division B		
5.	Ratio Analysis	Calculation of different ratios.
6.	Cost of Capital	Calculation of different cost of capitals.
7.	Capital Structure	Calculation of market value.
8.	Leverage	Calculation of leverages and EPS.
9.	Investment Decision	Calculation of NPV, MIRR, Payback period, ARR and Desirability Factor.
10.	Dividend Decision	Analysis of dividend policy using Gordon's and Walter's model.

11(a)	Sources of Finance	Features of Bulldog Bond.
11(b)	Sources of Finance	Spontaneous sources of finance.
11(c)	Capital Structure	Causes of over – capitalization.
11(d)	Scope and Objectives of Financial Management	Disadvantages of profit maximisation.

Section – B: Strategic Management

The Revision Test Paper on the subject of Strategic Management for the January, 2026 examination contains sixteen questions. The questions have been selected from all the sections/chapters uniformly to cover whole syllabus. Questions are based on different skill levels, i.e., "Comprehension & Knowledge" as well as "Analysis & Application".

The questions included are of different categories – multiple choice questions based on case scenario and application based multiple choice questions, distinguish between, short notes, descriptive and questions based on practical scenarios. The first question contains multiple choice questions based on case scenario are subdivided into five different parts. Questions from two to six are application based multiple choice questions. All multiple-choice questions are given with four alternatives and the student has to opt for the correct option. Subsequently, ten different questions have been included to cover all the five chapters of the syllabus. Chapter names have been mentioned before questions. A descriptive question based on practical scenario has been included from each section. Another descriptive question has also been included from each section of the syllabus.

The students should take up this Revision test paper as a tool to check their preparedness in the subject. Mere reading of Revision Test Paper will not be helpful. To properly self-assess the preparation in the subject, students must attempt the questions on their own. Compare your answers with the suggested answers and hints given to assess the level of preparation and identify areas where more focus is required. Then you may work on these areas to improve the quality of answers that you write.

Work hard and perform well in the examination!

PAPER – 4: COST AND MANAGEMENT ACCOUNTING

QUESTIONS

Case Scenario 1

EcoTrans Logistics Pvt. Ltd. is a regional goods transport company based in the industrial hub of LogiPort, providing daily freight services to and from GreenCity, a fast-growing metropolitan area located 120 kilometers away. The company has been operating for over a decade and has built a reputation for timely deliveries, especially for fast-moving consumer goods (FMCGs), packaged food items, and light industrial components.

To meet increasing demand, EcoTrans has recently expanded its fleet and now operates 24 transport vehicles of varying capacities. The company promises reliable daily service, and each vehicle typically completes one round trip per day, carrying cargo in both directions. EcoTrans emphasizes sustainability and efficiency, with systems in place for fuel monitoring, vehicle maintenance, and cost control.

However, to remain competitive and ensure profitability in a cost-sensitive industry, management has decided to undertake a detailed cost analysis for the month of April 2025. This analysis will guide pricing strategy and assess whether current operations are yielding the desired profit margin of 20% on freight revenue.

The following data has been compiled by the Finance and Operations departments for cost analysis and pricing decision-making.

Fleet Composition:

No. of Vehicles	Capacity per Vehicle (MT)
6	10 MT
8	14 MT
6	18 MT
4	22 MT

Each vehicle operates one round trip per day on average. Vehicles are loaded to 85% capacity on the forward journey and 65% on the return. On average, 12% of the fleet is under maintenance at any time.

The company operated for 26 days in April 2025.

Monthly Expenditure (April 2025)

Description	Amount (₹)
Salary of Transport Manager	75,000
Salary of 32 Drivers @ ₹22,000 each	7,04,000
Wages of 28 Helpers @ ₹13,000 each	3,64,000
Loading & Unloading Charges per Trip	950
Consumable Stores (Fuel Additives, Tools)	1,50,000
Toll Charges	250 per Trip
Insurance (Annual Premium)	9,60,000
Road Licence Fee (Annual)	5,40,000
Diesel Cost per Litre	82
Mileage per Vehicle	4.5 km/litre
Lubricants & Oils	1,30,000
Tyres, Tubes, Spare Parts (Running basis)	4,80,000
GPS & Fleet Tracking Subscription (Monthly Basis)	2,000 per vehicle
Driver Safety & Training Programmes	80,000
Garage Rent (Annual)	10,80,000
Workshop Repairs (Routine & Emergency)	3,40,000

Electricity & Utility Expenses	62,000
Depreciation on Vehicles (Monthly)	6,80,000

The company operates an in-house workshop that handles routine and emergency repairs for both its own fleet and external vehicles. As part of the internal cost allocation, 45% of the Transport Manager's salary is charged to the workshop to reflect time and resources dedicated to overseeing maintenance operations. In return, the transport department has been apportioned ₹95,000 by the workshop for repair and service support during the month.

Being a finance manager of the company, you are required to answer the followings (5 MCQs):

(i) What is the total fixed cost for April 2025?

(A) ₹21,95,000

(B) ₹20,89,750

(C) ₹22,89,250

(D) ₹19,98,250

(ii) Calculate the total variable cost for April 2025

(A) ₹44,71,885

(B) ₹31,50,000

(C) ₹29,65,920

(D) ₹48,45,000

(iii) What should be the freight rate per ton-km to achieve a 20% profit margin?

(A) ₹4.461

(B) ₹5.576

(C) ₹4.372

(D) ₹3.845

(iv) Calculate the total wages paid to loading and unloading labour in April 2025.

(A) ₹10,43,520

(B) ₹10,42,560

(C) ₹9,85,000

(D) ₹10,37,400

(v) What is the total ton-kilometers (ton-km) generated by EcoTrans Logistics in April 2025?

(A) 1,31,789 ton-km

(B) 5,49,120 ton-km

(C) 15,15,571 ton-km

(D) 26,00,000 ton-km

Case Scenario 2

In order to provide employment opportunities in rural areas of Madhya Pradesh, State government decided to give interest free loan up to ₹ 30,00,00,000 to Khadi cloth industrialists. In return these industrialists are expected to set up the factory and produce khadi cloth in such rural areas.

The loan is refundable after 20 years in lump sum. Madhya Pradesh government kept a stipulation that 40% of cloth production shall be supplied to government at a nominal rate of ₹ 10 per metre. This supply shall be used by the government for making uniforms of class IV employees of state government.

Haryana cloth mill of Hissar took loan of ₹ 30 crores and commenced business in Dindori, a rural district of Madhya Pradesh. Production commenced at Dindori and following are the details:

Production during the year 2023-24	**40,00,000 metres of Khadi cloth**
Raw cotton used	₹ 1,20,00,000
Direct wages	₹ 1,80,00,000
Freight inwards	₹ 8,00,000
Factory overheads estimated at 30% of prime cost	
Office and administration overheads estimated at 20% of factory cost	

Donations	₹ 2,00,000
Fines & Penalties	₹ 5,00,000
Selling and distribution overhead estimated at	₹ 1,69,52,000

The selling price to be charged from general public is to be fixed in such a way as to earn a profit of 30% on total cost of sales and price should be in multiple of Re.1.

Actual overheads incurred are as under:-

(1) Factory overheads = ₹ 89,50,000

(2) Office and administration overheads = ₹ 90,50,000

(3) Selling and distribution overheads = ₹ 1,70,00,000

The company has been requested to submit a cost audit report to the government detailing the production activities in its factory. As part of this process, the government has provided a draft format containing several key questions that must be addressed within the report. These questions are designed to ensure the accuracy and transparency of the company's production costs, and they play a critical role in shaping the audit report. Before proceeding with the preparation of the final cost audit report, it is essential that the company carefully reviews and works out the answers to these questions.

(i) Determine cost of sales per metre of cloth produced?

(A) ₹ 13.25

(B) ₹ 16.25

(C) ₹ 18.25

(D) ₹ 15.25

(ii) Determine the loss element per metre in government supply?

(A) ₹ 7.25

(B) ₹ 5.75

(C) ₹ 4.50

(D) ₹ 6.25

(iii) What will be the selling price per metre charged from general public?

(A) ₹ 29

(B) ₹ 27

(C) ₹ 31

(D) ₹ 33

(iv) What will be the rate of return on overall sales proceeds as per financial accounts?

(A) 22.313%

(B) 18.781%

(C) 24.387%

(D) 20.564%

(v) What will be the net profit in financial accounts from the factory, if government fixed the price of the cloth to be sold to general public at ₹ 28?

(A) ₹ 1,67,20,000

(B) ₹ 1,66,00,000

(C) ₹ 1,66,20,000

(D) ₹ 1,67,00,000

3. A manufacturing company has set the standard cost for producing one unit of its product as follows:

- Direct Material: 10 kg @ ₹ 5/kg
- Direct Labour: 4 hours @ ₹ 20/hour
- Variable Overhead: 4 hours @ ₹ 5/hour
- Fixed Overhead: ₹ 10 per unit (based on normal output of 5,000 units)

During a particular month, the company produced 4,800 units. The actual results were:

- Direct Material Used: 49,000 kg costing ₹ 2,45,000

- Direct Labour: 19,200 hours costing ₹ 3,84,000
- Variable Overhead Incurred: ₹ 95,000
- Fixed Overhead Incurred: ₹ 52,000

Based on this information, which of the following statements is most accurate?

(A) The material usage variance is ₹ 5,000 (Favourable)

(B) The labour rate variance is ₹ 19,200 (Adverse)

(C) The labour rate variance is ₹ 19,200 (Favourable)

(D) The fixed overhead volume variance is ₹ 2,000 (Adverse)

4. A manufacturing firm has produced 10000 units of a standard product by incurring the following expenses:

Material costs	:	₹ 80,000
Labour costs	:	₹ 40,000
Production overhead	:	₹ 30,000
Office overhead	:	20% of factory cost

Estimated selling and distribution overhead is ₹ 4 per unit. If the firm has no inventory at the beginning of the period and expects to maintain 20% of output as closing inventory, at what price per unit the product should be sold to earn a profit of 20% on sales?

(A) ₹ 27.50

(B) ₹ 23.00

(C) ₹ 26.40

(D) ₹ 28.75

5. Product AB has a standard cost of ₹ 32, ₹ 6 of which relates to direct materials. Budgeted production for the month was 1600 units. During the month, 1500 units of AB were produced and materials worth ₹ 9,000 were purchased. There was no opening stock of materials but closing stock, which was valued at standard cost, amounted to ₹ 1,500. What is the total variance for materials?

(A) ₹ 500 (F)

(B) ₹ 500 (A)

(C) ₹ 1500 (F)

(D) ₹ 1500 (A)

6. Rinki Ltd. has two divisions, A and B. Target profit of A is ₹ 100 lakh. A has capacity to produce 20,000 units of product P but only 15,000 units are produced and sold in the market at ₹ 3,000 per unit. B received an order for which P would be required as input. B approaches A for purchase of 5000 units of P. What price should A charge from B per unit of P so as to meet its target profit? (Other details of A are: Fixed cost is ₹ 90 lakh; Variable cost per unit of P is ₹ 2,000)

(A) ₹ 3,000

(B) ₹ 2,800

(C) ₹ 2,500

(D) ₹ 2,000

7. A company plans to produce 44,000 units during the budget period. The company requires 4 units of raw material to produce 1 unit of finished goods. The company currently has 1,00,000 units of raw material on hand and wishes to have an inventory of 1,10,000 units of raw material on hand at the end of the budget period. The number of raw material units to be purchased during the budget period is

(A) 1,76,000 units

(B) 1,67,000 units

(C) 1,86,000 units

(D) 1,68,000 units

Material Cost

8. XYZ Enterprises manufactures different types of beverages. The monthly demand pattern for these beverages is as follows:

- Orange Juice: 60,000 litres
- Apple Juice: 20,000 litres
- Grape Juice: 5,000 litres

To produce one litre of Apple Juice, Orange Juice, and Grape Juice, 6 kg of apples, 3 kg of oranges, and 5 kg of grapes are required, respectively. There is no opening or closing stock of raw materials.

XYZ Enterprises can source the raw materials either from local farmers (import in case or grapes) or from wholesale suppliers. The company has the flexibility to purchase any quantity of materials from the wholesale suppliers, but when buying from the farmers or in case of import, it must adhere to a minimum quantity that is specified for each type of material. Below is a summary of the purchasing conditions:

Material	Wholesale Supplier	Farmers
Oranges	Minimum purchase: Any quantity	7,20,000 kg
	Price per kg: ₹ 18.00	Price per kg: ₹ 15.00
	Transportation: ₹ 5,000	Transportation: ₹ 20,000
	-	Sorting and piling cost: ₹ 1,500
	Loading cost: ₹ 10/50 kg	Loading cost: ₹5/50 kg
	Unloading cost: ₹ 2/50 kg	Unloading cost: ₹2/50 kg
Apples	Minimum purchase: Any quantity	3,60,000 kg
	Price per kg: ₹ 12.00	Price per kg: ₹ 10.00
	Transportation: ₹ 11,000	Transportation: ₹ 15,000
	-	Sorting and piling cost: ₹ 1,200
	Loading cost: ₹10/50 kg	Loading cost: ₹ 3/50 kg
	Unloading cost: ₹2/50 kg	Unloading cost: ₹ 2/50 kg
	Wholesale Supplier	**Import**
Grapes	Minimum purchase: Any quantity	1,50,000 kg
	Price per kg: ₹ 32.00	Price per kg: ₹ 25.00
	Transportation: ₹ 3,000	Transportation: ₹ 15,000
	Loading cost: ₹ 10/50 kg	Loading cost: ₹ 25/50 kg
	Unloading cost: ₹2/50 kg	Unloading cost: ₹2/50 kg

Transportation and Sorting and piling cost is given for per purchase order.

8% import duty is applicable is purchase of grapes for which ITC is not available.

XYZ Enterprises is also paying an annual interest rate of 15% on cash credit facility and ₹1,000 per 1,000 kg of rent to its warehouse.

You are required to:

1. Calculate the total purchase cost for each material (Apple, Orange, and Grape) under the following two options:
   (a) Purchase from Wholesale Supplier
   (b) Purchase from Farmers/Import
2. Calculate the Economic Order Quantity (EOQ) for each material under both options.
3. Provide a Recommendation on the best purchase option for the "Grapes" material if minimum purchase quantity for import is 1,50,000 kg.

Employee Cost

9. Tommy HL Ltd. pays the following to a skilled worker engaged in production works. The following are the employee benefits paid to the employee:

(a)	Basic salary per day	₹1,000
(b)	Dearness allowance (DA)	20% of basic salary
(c)	House rent allowance	16% of basic salary
(d)	Transport allowance	₹50 per day of actual work
(e)	Overtime	Twice the hourly rate (considers basic and DA), only if works more than 9 hours a day otherwise no overtime allowance. If works for more than 9 hours a day then overtime is considered after 8th hours.

(f)	Work of holiday and Sunday	Double of per day basic rate provided works atleast 4 hours. The holiday and Sunday basic is eligible for all allowances and statutory deductions.
(h)	Earned leave & Casual leave	These are paid leave.
(h)	Employer's contribution to Provident fund	12% of basic and DA
(i)	Employer's contribution to Pension fund	7% of basic and DA

The company normally works 8-hour a day and 26-day in a month. The company provides 30 minutes lunch break in between.

During the month of August 2025, Mr. Shyam works for 23 days including 15th August and a Sunday and applied for 3 days of casual leave. On 15th August and Sunday he worked for 5 and 6 hours respectively without lunch break.

On 5th and 13th August he worked for 10 and 9 hours respectively.

During the month Mr. Shyam worked for 100 hours on Job no. AB100.

You are required to CALCULATE:

(i) Earnings per day

(ii) Effective wages rate per hour of Mr. Shyam.

(iii) Wages to be charged to Job no. AB100.

Overheads: Absorption Costing method

10. BrightTech Appliances Ltd. is a leading manufacturer of smart home devices such as the SmartCool AC, PowerLite Inverter, and EcoDry Dryer. Amid increasing operational complexity, the company's finance controller has initiated a review of overhead allocations to gain better control over departmental efficiency and product profitability.

The company operates through four major departments:

1. **Production** – handles manufacturing across three plants using advanced machinery including stores management.

2. **Administration** – supports HR, finance, legal, and compliance.
3. **Sales & Distribution** – manages retail tie-ups, e-commerce, and logistics.
4. **General Management** – oversees strategy, innovation, and executive-level operations.

The extract from the budget for the next financial year is as follows:

Particulars	Amount (₹)	Notes
Raw Material Cost	5,75,00,000	Consumed in a ratio of 3:4:3 for the three products
Indirect Material Cost	12,50,000	Manufacturing ₹7,20,000, Stores ₹25,000, General admin ₹3,80,000, Personnel ₹95,000, Sales ₹30,000
Salary & Wages	4,20,00,000	Includes direct wages of ₹1,60,00,000 (temporary contract workers)
Rent & Property Tax	₹1,20,000	₹90,000 Warehouse Rent, ₹30,000 Property Tax
Depreciation on Non-Current Assets	₹25,00,000	Factory/office building ₹10,50,000, ACs ₹2,00,000, Machinery ₹12,50,000
Power & Fuel	₹4,90,000	₹4,70,000 for manufacturing, ₹20,000 for delivery vans
Machinery Insurance Premium	₹5,00,000	At 4% of WDV of machinery
Group Employee Insurance	₹3,10,000	Based on gross salary (excluding direct workers and top management)
Printing & Stationery	₹8,20,000	Manufacturing ₹21,000, Finance ₹5,50,000, Legal ₹28,000, Sales ₹2,21,000
Audit Fees	₹1,40,000	Statutory and internal audit

Electricity Expense	₹4,00,000	Metered use: Production 5,600, Admin 10,000, Sales & Distribution 4,000, General Management 1,600 units
Telephone & Mobile	₹4,95,000	Production ₹1,25,000, Personnel ₹50,000, General Mgmt ₹30,000, Sales ₹75,000, Customer Support ₹2,15,000
Travelling Expenses	₹24,00,000	Production ₹5,50,000, General Mgmt ₹12,00,000, Sales ₹6,50,000
Meal Coupon Subsidy	₹2,25,000	₹3,000 per employee on roll
Software License Renewals	₹16,50,000	Stores Management ₹8,50,000, Finance ₹1,50,000, Stores ₹30,000, Customer Support ₹6,20,000
Misc. Expenditures	₹9,50,000	Allocated based on direct expenses

Additional Information:

- **Employee Count:** Production 22, Admin 20, Sales & Distribution 28
- **Average Gross Salary:** Production ₹6,30,000, Admin ₹4,20,000, Sales & Distribution ₹3,50,000, General Mgmt ₹2,80,000
- **Floor Area:** Production 9,900 m^2, Admin 3,600 m^2, Sales & Distribution 2,700 m^2, General Mgmt 1,800 m^2
- **AC Tonnage:** Production 6,000 RT, Admin 3,000 RT, S&D 3,000 RT, General Mgmt 1,500 RT
- **Depreciation Rates:** Building 5%, AC 15%, Machinery 10%
- **Department Roles:**
  - General Mgmt: 70% Sales strategy, 20% Production/ Marketing, 10% Admin
  - Admin: 50% Production, 50% Sales.

Required:

(i) Prepare a schedule of Cost Allocation for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(ii) Prepare a schedule of Cost Apportionment (Primary Distribution) for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(iii) Prepare a schedule of Secondary distribution of Administration Dept. and General Management costs.

Activity Based Costing

11. Edward Ltd. manufactures weighing machines of standard size and sells its products to two industrial customers namely MT Ltd. and KG Ltd. and to a dealer MG Bros. having shops in different cities. The maximum retail price per unit of weighing machine is ₹ 11,000 and per unit average cost of production is ₹ 5,500 (40% is general fixed overhead cost).

The Finance Officer has been asked to undertake a customer profitability analysis and calculate and compare the profit margin per customer (before deducting general fixed overhead) to know about the real customer profitability.

Following are the additional overhead information:

Delivery costs	₹ 200 per kilometer
Emergency delivery cost (in addition to delivery cost)	₹ 21,000 per delivery
Order processing cost	₹ 6,000 per order
Specific discount and sales commission	As per negotiation
Product Advertisement cost	Actual cost

The following data are available for each customer.

	MT Ltd.	KG Ltd.	MG Bros.
Sales (in units)	2,000	1,000	800
Total delivery kilometer travelled	1,000	800	900
No. of emergency delivery	2	1	0
No. of orders processed	4	2	8
Specific Discount (percentage of sales revenue)	25%	20%	15%
Sales commission (percentage of sales revenue)	15%	10%	5%
Advertisement costs (₹)	8,75,000	6,15,000	4,30,000

You are required to analyse the profitability for each customer, which customer is the most profitable.

Cost Accounting Systems

12. XYZ Limited manufactures a standard product in a competitive market. During the year, the company experienced operational changes, including partial automation and fluctuations in input costs. Although the company continued using traditional costing methods, discrepancies were observed between financial and cost profits. Management has requested a detailed analysis to understand the variance and assess the accuracy of overhead absorption.

The following is the Trading and Profit & Loss Account of XYZ Limited:

Dr.	₹	Cr.	₹	₹
To Raw Materials	32,50,000	By Sales (40,000 units)	92,00,000	
To Direct Wages	21,75,000	By Closing Finished Goods (2,000 units)	3,90,000	
To Production Overheads	11,20,000	By Work-in-Progress:		
To Administration Overheads	6,25,000	Materials	72,000	

To Selling & Distribution Overheads	4,60,000	Wages	34,000	
To Preliminary Expenses Written Off	30,000	Production Overheads	26,500	1,32,500
To Goodwill Written Off	55,000	By Dividend Received	5,25,000	
To Fines	10,000	By Interest from Bank Deposits	1,20,000	
To Interest on Term Loan	18,000			
To Loss on Sale of Equipment	20,000			
To Tax	2,25,000			
To Net Profit for the Year	23,79,500			
Total	1,03,67,500		Total	1,03,67,500

XYZ Limited manufactures a standard product and uses traditional costing systems for internal reporting.

The Cost Accounting records of XYZ Ltd. show the following:

1. Production Overheads have been absorbed into work-in-progress at 25% on Prime Cost.
2. Administration Overheads have been charged at ₹14.75 per unit of finished goods produced.
3. Selling and Distribution Overheads have been charged at ₹11.50 per unit sold.
4. The under- or over-absorption of overheads has not been adjusted in the Costing Profit & Loss Account.

You are required to:

1. Prepare a Proforma Costing Profit and Loss Account, showing the net profit as per cost accounts.

2. Prepare control accounts for:
   - Production Overheads
   - Administration Overheads
   - Selling & Distribution Overheads
3. Reconcile the profit as per cost accounts with the profit shown in the financial accounts.

Job & Batch Costing

13. A manufacturing company follows a job costing system. The following data is extracted from its books for the year ended 31st March, 2025:

Particulars	Amount (₹)
Direct Materials	12,50,000
Direct Wages	9,80,000
Selling & Distribution Overheads (Variable: 60%, Fixed: 40%)	6,30,000
Administration Overheads (Fixed)	5,40,000
Factory Overheads (70% Fixed, 30% Variable)	5,75,000
Profit	8,25,000

Additional Information:

1. **Overhead Recovery:**
   - Factory overheads are absorbed as a % of Direct Wages.
   - Selling & Distribution and Administration overheads are recovered as a % of Cost of Production.
2. In 2025-26, the company received a new job order. It is estimated that direct materials required will be ₹ 3,20,000 and direct labour will cost ₹ 2,10,000

Additional Information:

- Selling & Distribution overheads have increased by 20% (with variable portion now being 65%).
- Factory overheads have decreased by 10% due to efficiency improvements.

- The company wants to maintain the same profit % on sales as in the previous year.

You are required to prepare:

(a) Prepare a Job Cost Sheet showing Prime Cost, Cost of Production, Cost of Sales, Sales Value

(b) Determine the price to be quoted for the new job in 2025-26, considering Revised overhead structure and the company's profit objective.

(Assume relevant cost rates are based on previous year's data unless specified otherwise.)

Process Costing

14. Following data are available for a product for the month of April, 2025:

Particulars	Process- I (₹)	Process- II (₹)
Opening work-in- progress	Nil	Nil
Costs incurred during the month:		
- Direct materials	6,00,000	
- Labour	1,20,000	1,60,000
- Factory overheads	2,40,000	2,00,000
Units of production:		
Received in process	40,000	36,000
Completed and transferred	36,000	32,000
Closing work-in-progress	2,000	?
Normal loss in process	2,000	1,500

Production remaining in process has to be valued as follows:

Materials 100% Labour 50% Overheads 50%

There has been no abnormal loss in Process- II.

The company follows weighted average method for valuing inventory.

Prepare Process Accounts after working out the missing figures and with detailed workings.

Joint Product and By-Product

15. A company processes a raw material in its Department 1 to produce three products, viz. P, Q and C at the same split-off stage. During a period 1,80,000 kgs of raw materials were processed in Department 1 at a total cost of ₹ 12,88,000 and the resultant output of P, Q and C were 18,000 kgs, 10,000 kgs and 54,000 kgs respectively. P and Q were further processed in Department 2 at a cost of ₹1,80,000 and ₹1,50,000 respectively.

    C was further processed in Department 3 at a cost of ₹1,08,000. There is no waste in further processing. The details of sales affected during the period were as under:

	P	Q	C
Quantity Sold (kgs.)	17,000	5,000	44,000
Sales Value (₹)	12,24,000	2,50,000	7,92,000

    There were no opening stocks. If these products were sold at split-off stage, the selling prices of P, Q and C would have been ₹ 50, ₹ 40 and ₹ 10 per kg respectively.

    Required:

    (i) Prepare a statement showing the apportionment of joint costs to P, Q and C.

    (ii) Present a statement showing the cost per kg of each product indicating joint cost and further processing cost and total cost separately.

    (iii) Prepare a statement showing the product wise and total profit for the period.

    (iv) State with supporting calculations as to whether any or all the products should be further processed or not

Standard Costing

16. A company produces a product X, using raw materials A and B. The standard mix of A and B is 1: 1 and the standard loss is 10% of input. You are required to COMPUTE the MISSING INFORMATION indicated by "?" based on the data given below:

	A	B	Total
Standard price of raw material (₹ / kg.)	24	30	
Actual input (kg.)	?	70	
Actual output (kg.)			?
Actual price ₹ / kg.	30	?	
Standard input quantity (kg.)	?	?	
Yield variance (sub usage)	?	?	270(A)
Mix variance	?	?	?
Usage variance	?	?	?
Price variance	?	?	?
Cost variance	0	?	1,300(A)

Marginal Costing

17. Jupiter Ltd, a 'Fast-Moving Consumer Goods (FMCG)' company intends to diversify the product line to achieve full utilisation of its plant capacity. As a result of considerable research made, the company has been able to develop a new product called 'EXE'.

'EXE' is packed in *cans* of 100 ml capacity and is sold to the wholesalers in cartons of 24 *cans* at ₹120 per carton. Since the company uses its spare capacity for the manufacture of 'EXE', no additional fixed expenses will be incurred. However, accountant has allocated a share of ₹1,12,500 per month as fixed expenses to be absorbed by 'EXE' as a fair share of the company's present fixed costs to the new product for costing purposes.

The company estimates the production and sale of 'EXE' at 1,50,000 *cans* per month and on this basis the following cost estimates (per carton) have been developed:

	₹
Direct Materials	54
Direct Wages	36
All Overheads	27
Total Costs	117

After a detailed market survey the economy is confident that the production and sales of 'EXE' can be increased to 1,75,000 *cans* per month and ultimately to 2,25,000 *cans* per month.

The company at present has a capacity for the manufacture of 1,50,000 empty *cans* and the cost of the empty *cans* if purchased from outside will result in a saving of 20% in direct material, 10% in direct wages and 10% in variable overhead costs of 'EXE'. The price at which the outside firm is willing to supply the empty *cans* is ₹0.675 per empty *can*. If the company desires to manufacture empty *cans* in excess of 1,50,000 *cans*, a machine involving an additional fixed overhead of ₹7,500 per month will have to be installed.

Required:

(a) State by showing your workings whether the company should make or buy the empty *cans* at each of the three volumes of production of 'EXE' namely, 1,50,000, 1,75,000 and 2,25,000 *cans*.

(b) At what volume of sales will it be economical for the company to install the additional equipment for the manufacture of empty cans?

(c) Evaluate the profitability on the sale of 'EXE' at each of the aforesaid three levels of output based on your decision and showing the cost of empty *cans* as a separate element of cost.

Budget and Budgetary Control

18. SIAM Ltd. manufactures two products using one type of material and one grade of labour. Shown below is an extract from the company's working papers for the next period's budget.

Particulars	**Product A**	**Product B**
Budgeted Sales (Units)	1,800	2,400

Budgeted Material Consumption per Product (Kg.)	5	3
[Budgeted Material Cost ₹12 per Kg.]		
Standard Hours Allowed per product	5	4
[Budgeted Wage Rate ₹8 per hour]		

Overtime premium is 50% and is payable, if a worker works for more than 40 hours a week. There are 45 direct workers.

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%; in addition the non-productive downtime is budgeted at 20% of the productive hours worked.

There are twelve 5-day weeks in the budget period and it is anticipated that sales and production will occur evenly throughout the whole period.

It is anticipated that stock at the beginning of the period will be:

Product A 510 units; Product B 1,200 units; Raw material 2,150 Kg.

The target closing stock, expressed in terms of anticipated activity during budget period are - Product *A* 15 days sales; Product *B* 20 days sales; Raw material 10 days consumption.

Required:

(a) the material purchases budget, and

(b) the wages budget for the direct workers, showing the quantities and values, for the next period.

Miscellaneous

19. (a) Standard costs, marginal costs are some of the types of cost which helps the management in decision making. DISCUSS any five types of costs categorised based on its use in Managerial Decision Making.

(b) Provide EXAMPLE(S) of the cost driver for the following cost pools:

Cost Pool
Quality Control

Research and Development
Machine Maintenance
Employee Training Costs
Customer Service

(c) Besides having advantages of Budgetary Control System being a powerful instrument used by business entity for the control of their expenditure, it has certain limitations as well. ELABORATE any four limitations of Budgetary Control System.

(d) Cost control emphasis is on past and present, while cost reduction emphasis is on present and future. DISCUSS some more differences between Cost control and Cost reduction.

SUGGESTED ANSWERS/HINTS

1. (i) (C) ₹22,89,250

(ii) (A) ₹44,71,885

(iii) (B) ₹5.576

(iv) (D) ₹10,37,400

(v) (C) 15,15,571 ton-km

Operating Cost Sheet for the month

Particulars		**Amount (₹)**
A.	**Fixed Charges:**	
	Transport Manager's Salary (55% x ₹75,000)	41,250
	Driver Salaries (32 x ₹22,000)	7,04,000
	Helper Wages (28 x ₹13,000)	3,64,000
	GPS Subscription (24 x ₹2,000)	48,000
	Driver Safety & Training Programmes	80,000
	Insurance (₹9,60,000 ÷ 12)	80,000

	Road Licence Fee (₹5,40,000 ÷ 12)	45,000
	Garage Rent (₹10,80,000 ÷ 12)	90,000
	Electricity & Utility Expenses	62,000
	Depreciation on Vehicles	6,80,000
	Workshop Repairs (Apportioned)	95,000
	Total Fixed Costs	**22,89,250**
B.	**Variable Charges:**	
	Loading & Unloading Charges (WN.1)	10,37,400
	Toll Charges (W.N.2)	2,73,000
	Diesel Cost (W.N.3)	24,01,485
	Lubricants & Oils	1,30,000
	Tyres, Tubes, Spare Parts	4,80,000
	Consumable Stores	1,50,000
	Total Variable Costs	**44,71,885**
C.	**Total Cost (A + B)**	**67,61,135**
D.	Total Ton-Kms. (W. N. 4)	15,15,571
E.	**Cost per ton-km. (C ÷ D)**	**4.461**

Calculation of Chargeable Freight

Cost per ton-km.	₹ 4.461
Add: Profit @ 20% on freight or 25% on cost	₹ 1.115
Chargeable freight per ton-km.	**₹ 5.576**

Working Notes:

1. Wages paid to loading and unloading labours

Numbers of vehicles available per day x No. of days x trips x wages per trip

(24 vehicles x 88%) x 26 days x 2 trips x ₹950

21 x 2 x 26x 950 = ₹10,37,400

2. Toll Charges

Numbers of vehicles available per day x No. of days x trips x toll charges per trip

21 x 2 x 26x 250 = ₹2,73,000

3. Cost of Diesel:

Distance covered by each vehicle during the month

= 120 k.m. × 2 × 26 days × 88% = 5,491.2 km.

Consumption of diesel = $\frac{5,491.2\text{k.m.} \times 24\,\text{vehicles}}{4.5\text{k.m.}}$ = 29,286.4 litres

Cost of diesel = 29,286.4 litres × ₹ 82 = ₹24,01,485

4. Calculation of total ton-km:

Total Ton-Km. = Total Capacity × Distance covered by each vehicle × Average Capacity Utilisation ratio.

$$= \left[(6 \times 10\,\text{MT}) + (8 \times 14\text{MT}) + (6 \times 18\,\text{MT}) + (4 \times 22\,\text{MT})\right] \times 5,491.2\text{k.m.} \times \frac{(85\% + 65\%)}{2}$$

= (60 + 112 + 108 + 88) x 5,491.2 x 75%

= 368 x 5,491.2 x 75%

= 15,15,571 ton-km.

2. (i) (B) ₹ 16.25

Particulars	**Amount (₹)**	**Amount (₹)**
Raw material	1,20,00,000	
Add: Freight inwards	8,00,000	
Direct material:		1,28,00,000
Direct labour (wages)		1,80,00,000
Prime cost		**3,08,00,000**
Add: Factory overhead 30% of 3,08,00,000		92,40,000
Factory cost		**4,00,40,000**

Add: Office and administration overheads 20% of 4,00,40,000		80,08,000
Cost of production		**4,80,48,000**
Add: Selling and distribution overheads		1,69,52,000
Cost of sales		**6,50,00,000**
Add: Profit 30% of 6,50,00,000		1,95,00,000
Desirable sales		**8,45,00,000**

$$\text{Cost of sales per metre} = \frac{₹\ 6{,}50{,}00{,}000}{40{,}00{,}000 \text{ metres}} = ₹\ 16.25 \text{ per metre}$$

(ii) (D) ₹ 6.25

Loss element per metre on government supply

Cost of sales of khadi cloth per metre	₹ 16.25
Less: Supply rate to government per metre	(₹ 10)
Loss element per metre	₹ 6.25

(iii) (A) ₹ 29

Particulars	
Desirable sale as per cost sheet	₹ 8,45,00,000
Less: Amount at which supply is made to MP govt. (40,00,000 m x 40% x ₹ 10)	(₹ 1,60,00,000)
Amount to be recouped from general public	₹ 6,85,00,000
Units to be sold to general public	24,00,000 m
Selling price to general public $\frac{₹\ 6{,}85{,}00{,}000}{24{,}00{,}000\ \text{m}}$	₹ 28.5416

This will be rounded-off to Re. 1, so selling price will be ₹ 29 per metre.

(iv) (A) 22.313%

Profit & loss account

Particulars	**Amount (₹)**	**Particulars**	**Amount (₹)**
To direct material	1,20,00,000	By sales (₹ 29 x 24,00,000 + 10 x 16,00,000)	8,56,00,000
To direct labour	1,80,00,000		
To freight inwards	8,00,000		
To factory overheads	89,50,000		
To office and administration	90,50,000		
To selling and distribution overhead	1,70,00,000		
To Donations	2,00,000		
To fines and penalties	5,00,000		
To profit	1,91,00,000		
Total	**8,56,00,000**	**Total**	**8,56,00,000**

$$\text{Rate of return} = \frac{\text{Profit}}{\text{Sales}} = \frac{1{,}91{,}00{,}000}{8{,}56{,}00{,}000} = 22.313\%$$

(v) (D) ₹ 1,67,00,000

Particulars	**Amount (₹)**
New sales (₹ 28 x 24,00,000 + ₹ 10 x 16,00,000)	8,32,00,000
Less: Total costs as per P&L made above	(6,65,00,000)
New profit	**1,67,00,000**

3. (D) The fixed overhead volume variance is ₹ 2,000 (Adverse)

Material Usage Variance (MUV)

= (Standard Quantity - Actual Quantity) × Standard Price = (48,000-49,000) × ₹ 5 = (-1,000) × ₹ 5 = ₹ 5,000 (Adverse)

Labour Rate Variance (LRV)

= (Standard Rate - Actual Rate) × Actual Hours

= (₹ 20 - ₹ 20) × 19,200 = ₹ 0

Fixed Overhead Volume Variance

= (Actual Output – Budgeted Output) × Standard FOH rate per unit = (4,800 – 5,000) × ₹ 10 = ₹ 2,000 Adverse

4. (A) ₹ 27.50

Particulars	Amount (₹)
Material Costs	80,000
Labour Costs	40,000
Production Overheads	30,000
Factory Cost	**1,50,000**
Add: Office Overheads (20% of Factory Cost)	30,000
Cost of Production (10,000 units)	**1,80,000**
Less: Closing Stock (2,000 units @ ₹ 18)	(36,000)
Cost of Goods Sold (8,000 units)	**1,44,000**
Add: Selling & Distribution (8,000 × ₹ 4)	32,000
Total Cost of Sales	**1,76,000**
Add: Profit (20% of Sales = 44,000)	44,000
Sales Revenue	**2,20,000**

Selling Price per Unit = 2,20,000/8,000 = ₹ 27.50 per unit

5. (C) ₹ 1500 (F)

Material Cost Variance

= Standard cost of actual output – Actual cost of materials consumed

= (1,500 x 6) – (9,000 - 1,500)

= 1,500 (F)

6. (B) ₹ 2,800

Revenue = 15,000 × 3,000 = 450,00,000 (₹ 450 lakh)

Variable cost = 15,000 × 2,000 = 300,00,000 (₹ 300 lakh)

Contribution = 450 - 300 = ₹ 150 lakh

Profit = Contribution - Fixed cost = 150 - 90 = ₹ 60 lakh

Target Profit = 100 - 60 = ₹ 40 lakh

Required contribution per unit = 40,00,000/5,000 = ₹ 800/unit

Transfer price = Variable cost + 800 = 2,000 + 800 = ₹ 2,800/unit

7. (C) 1,86,000 units

Raw material required for production = 44,000 units x 4 units/unit = 1,76,000

Raw material to be purchased = Closing stock + RM required – Opening stock

= 1,10,000 + 1,76,000 – 1,00,000

= 1,86,000 units

8. 1. (i) Calculation of Purchase Cost per Kg. of Materials

	Wholesale Market (₹)	Farmers (₹)
Orange:		
Purchase price	18.00	15.00
Add: Loading Cost	0.20 (₹ 10 ÷ 50 Kg.)	0.10 (₹ 5 ÷ 50 Kg.)
Add: Unloading Cost	0.04 (₹ 2 ÷ 50 Kg.)	0.04 (₹ 2 ÷ 50 Kg.)
	18.24	15.14
Apple:		
Purchase price	12.00	10.00

Add: Loading Cost	0.20 (₹ 10 ÷ 50 Kg.)	0.06 (₹ 3 ÷ 50 Kg.)
Add: Unloading Cost	0.04 (₹ 2 ÷ 50 Kg.)	0.04 (₹ 2 ÷ 50 Kg.)
	12.24	20.10
Grape:		
Purchase price	32.00	25.00
Add: Import duty @ 8%	---	2.00
Add: Loading Cost	0.20 (₹ 10 ÷ 50 Kg.)	0.50 (₹ 25 ÷ 50 Kg.)
Add: Unloading Cost	0.04 (₹ 2 ÷ 50 Kg.)	0.04 (₹ 2 ÷ 50 Kg.)
	32.24	27.54

(ii) Economic Order Quantity (E.O.Q)

$$= \sqrt{\frac{2 \times \text{Annual requirement} \times \text{Ordering cost}}{\text{Carrying cost per kg. per annum}}}$$

Annual Requirement (A)

Commodity		**Quantity (Kg.)**
Orange	(60,000 Ltr. x 3 Kg. x 12 months)	21,60,000
Apple	(20,000 Ltr. x 6 Kg. x 12 months)	14,40,000
Grape	(5,000 Ltr. x 5 Kg. x 12 months)	3,00,000

Cost per Order (O)

	Wholesale Market (₹)	**Farmers (₹)**
Orange:		
- Transportation cost	5,000	20,000
- Sorting and piling cost	---	1,500
	5,000	21,500

Apple:		
- Transportation cost	11,000	15,000
- Sorting and piling cost	---	1,200
	11,000	16,200
Grape:		
- Transportation cost	3,000	15,000
- Sorting and piling cost	---	---
	3,000	15,000

Carrying Cost per Kg. per annum (C x i)

	Wholesale Market (₹)	**Farmers (₹)**
Orange:		
- Interest on cash credit	2.736 (₹ 18.24 x 15%)	2.271 (₹ 15.14 x 15%)
- Warehouse rent*	1.0000	1.0000
	3.736	3.271
Apple:		
- Interest on cash credit	1.836 (₹ 12.24 x 15%)	3.015 (₹ 20.10 x 15%)
- Warehouse rent	1.0000	1.0000
	2.836	4.015
Grape:		
- Interest on cash credit	4.836 (₹ 32.24 x 15%)	4.131 (₹ 27.54 x 15%)
- Warehouse rent	1.0000	1.0000
	5.836	5.131

* Warehouse rent per Kg. = $\frac{₹\ 1{,}000}{1{,}000\text{Kg.}}$ = ₹ 1

Calculation of E.O.Q for each material under the both options

	Wholesale Market (Kg.)	Farmers/Import (Kg.)
Orange	$\sqrt{\frac{2 \times 21,60,000\text{Kg.} \times ₹\ 5,000}{₹\ 3.736}}$ = 76,036.72	$\sqrt{\frac{2 \times 21,60,000\text{Kg.} \times ₹21,500}{₹\ 3.271}}$ = 1,68,508.11
Apple	$\sqrt{\frac{2 \times 14,40,000\text{Kg.} \times ₹11,000}{₹\ 2.836}}$ = 1,05,691.36	$\sqrt{\frac{2 \times 14,40,000\text{Kg.} \times ₹16,200}{₹\ 4.015}}$ = 1,07,798.07
Grape	$\sqrt{\frac{2\text{x}3,00,000\text{Kg.} \times ₹3,000}{₹\ 5.836}}$ = 17,562.19	$\sqrt{\frac{2 \times 3,00,000\text{Kg.} \times ₹15,000}{₹\ 5.131}}$ = 41,881.31

(iii) Selection of best purchase option for the purchase of Grapes

	Wholesale Market	Import
Annual Requirement (A) (Kg.)	3,00,000	3,00,000
Order Quantity (Q)	17,562.19	1,50,000
No. of orders $\left(\frac{A}{Q}\right)$	17	2
Average Inventory $\left(\frac{Q}{2}\right)$ (Kg.)	8,781	75,000
Ordering Cost (₹) (I)	51,000 (17 Orders x ₹ 3,000)	30,000 (2 Order x ₹ 15,000)
Carrying Cost (₹) (II) (Average Inventory x Carrying cost per kg.)	51,245.92 (8,781 Kg. x ₹ 5.836)	3,84,825 (75,000 Kg. x ₹ 5.131)
Purchase Cost (₹) (III)	96,72,000	82,62,000

	(3,00,000 Kg. x ₹ 32.24)	(3,00,000 Kg. x ₹ 27.54)
Total Cost (I) + (II) + (III)	97,74,245.92	86,76,825

Importing grapes is the best purchase option for the XYZ enterprises.

9. Workings

1. Normal working hours in a month = (Daily working hours – lunch break) × no. of days

   = (8 hours – 0.5 hours) × 26 days = 195 hours

2. Hours worked by Mr. Shyam = No. of normal days worked + Overtime + holiday / Sunday worked

   = (21 days × 7.5 hours) + (9.5 hours + 8.5 hours) + (5 hours + 6 hours)

   = 157.5 hours + 18 hours + 11 hours = 186.50 hours.

(i) Calculation of earnings per day

Particulars	**Amount (₹)**
Basic salary (₹ 1,000 × 26 days)	26,000
Dearness allowance (20% of basic salary)	5,200
	31,200
House rent allowance (16% of basic salary)	4,160
Employer's contribution to Provident fund (12% × ₹ 31,200)	3,744
Employer's contribution to Pension fund (7% × ₹ 31,200)	2,184
	41,288
No. of working days in a month (days)	26

Rate per day	1,588
Transport allowance per day	50
Earnings per day	**1,638**

(ii) Calculation of effective wage rate per hour of Mr. Shyam:

Particulars	**Amount (₹)**
Basic salary (₹ 1,000 × 26 days)	26,000
Additional basic salary for Sunday & holiday (₹ 1,000 × 2 days)	2,000
Dearness allowance (20% of basic salary)	5,600
	33,600
House rent allowance (16% of basic salary)	4,480
Transport allowance (₹ 50 × 23 days)	1,150
Overtime allowance (₹ 160 × 2 × 2 hours)*	640
Employer's contribution to Provident fund (12% × ₹33,600)	4,032
Employer's contribution to Pension fund (7% × ₹ 33,600)	2,352
Total monthly wages	46,254
Hours worked by Mr. Shyam (hours)	186.5
Effective wage rate per hour	**248**

*(Daily Basic + DA) ÷ 7.5 hours

= (1,000+200) ÷ 7.5 = ₹160 per hour

(iii) Calculation of wages to be charged to Job no. AB100

= ₹ 248 × 100 hours = ₹ 24,800

10. (i) Schedule of Cost Allocation among the Departments

	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Raw Material Cost	5,75,00,000	5,75,00,000	-	-	-
Indirect Material Cost:	12,50,000	7,45,000	4,75,000	30,000	-
Salary & Wages	1,60,00,000	1,60,00,000	-	-	-
Rent & Property tax	90,000	-	-	90,000	-
Depreciation on Machinery	12,50,000	12,50,000	-	-	-
Power & fuel	4,90,000	4,70,000	-	20,000	-
Insurance premium on machinery	5,00,000	5,00,000	-	-	-
Printing & Stationery:	8,20,000	21,000	5,78,000	2,21,000	-
Audit fees	1,40,000	-	-	-	1,40,000
Telephone & Mobile expenses:	4,95,000	1,25,000	50,000	2,90,000	30,000
Travelling expenses	24,00,000	5,50,000	-	6,50,000	12,00,000
Software license renewal fees:	16,50,000	8,80,000	1,50,000	6,20,000	-
Total Allocated Direct Expenses	8,25,85,000	7,80,41,000	12,53,000	19,21,000	13,70,000

(ii) Schedule of Cost Apportionment (Primary Distribution)

	Basis	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Allocated Cost	Direct	8,25,85,000	7,80,41,000	12,53,000	19,21,000	13,70,000
Salary & Wages	Gross Salary (9:6:5:4)	2,60,00,000	97,50,000	65,00,000	54,16,667	43,33,333
Rent & Property tax	Floor Area (11:4:3:2)	30,000	16,500	6,000	4,500	3,000
Depreciation						
– Building	Floor Area (11:4:3:2)	10,50,000	5,77,500	2,10,000	1,57,500	1,05,000
– AC	RT (4:2:2:1)	2,00,000	88,889	44,444	44,444	22,222

Group Employee Insurance	Gross Salary (9:6:5:4)	3,10,000	1,16,250	77,500	64,583	51,667
Electricity Expenses	Units (14:25:10:4)	4,00,000	1,05,660	1,88,679	75,472	30,189
Meal Coupon Subsidy	No. of Employees (22,20,28)	2,25,000	66,000	60,000	84,000	15,000 (b.f.)
Miscellaneous Expenditure	Direct Expenses (Allocated Expenses)	9,50,000	8,97,729	14,414	22,098	15,760
Total		11,17,50,000	8,96,59,528	83,54,037	77,90,264	59,46,170

(iii) Schedule of Secondary Distribution:

	Basis	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Total Allocated and Apportioned costs		11,17,50,000	8,96,59,528	83,54,037	77,90,264	59,46,170
General Management	(2:1:7)	59,46,170	11,89,234	5,94,617	41,62,319	(59,46,170)
			9,08,48,762	89,48,654	1,19,52,583	
Administration dept.	(1:1)	89,48,654	44,74,327	(89,48,654)	44,74,327	
Total			9,53,23,089	-	1,64,26,910	

11. Customer Profitability Statement

Particulars		MT Ltd.	KG Ltd.	MG Bros.
Sales (units)		2,000	1,000	800
		(₹)	(₹)	(₹)
Sales Revenue	(A)	2,20,00,000	1,10,00,000	88,00,000
Less: Average Variable Cost (₹ 5,500 × 60% = 3,300 p.u.)	(B)	66,00,000	33,00,000	26,40,000
Contribution [70%of Sales]	...(A)-(B)	1,54,00,000	77,00,000	61,60,000
Less: Additional Overheads				
Delivery Cost		2,00,000	1,60,000	1,80,000

(No. of K.M. × ₹ 200)			
Emergency Delivery Cost (Emergency Delivery × ₹ 21,000)	42,000	21,000	-
Order Processing Cost (No. of Orders × ₹ 6,000)	24,000	12,000	48,000
Specific Discount	55,00,000	22,00,000	13,20,000
Sales Commission	33,00,000	11,00,000	4,40,000
Advertisement Cost	8,75,000	6,15,000	4,30,000
Profit *per customer**	54,59,000	35,92,000	37,42,000
Profit Margin *per customer** (%)	24.81%	32.65%	42.52%
Rank	III	II	I

**Before deducting general fixed overhead cost*

The contribution margin is 70% for each customer but when the other overheads costs per customer is included in the above profitability statement the profitability of the three customers become different. MG Ltd. is the most profitable customer.

12. (i) Costing Profit & Loss Statement

Particulars	**Amount (₹)**
Raw Materials	32,50,000
Direct Wages	21,75,000
Prime Cost	**54,25,000**
Add: Production Overheads (25% of Prime Cost)	13,56,250
Factory Cost	**67,81,250**
Less: Work-in-Progress	(1,32,500)
Manufacturing Cost Incurred During the Period	**66,48,750**
Add: Administration Overheads (₹14.75 x 42,000 units)	6,19,500
Cost of Production	**72,68,250**
Less: Closing Finished Goods (2,000 units x ₹ 173.05)	(3,46,100)
Cost of Goods Sold	**69,22,150**

Add: Selling & Distribution Overheads (₹11.50 x 40,000 units)	4,60,000
Cost of Sales	**73,82,150**
Profit (Balancing Figure)	18,17,850
Sales (40,000 units)	**92,00,000**

(ii) Control Accounts

Production Overheads A/c

Particulars	Amount (₹)	Particulars	Amount (₹)
To General Ledger Adjustment A/c	11,20,000	By WIP A/c (absorbed)	13,56,250
To Overhead Adjustment A/c (Over-absorption)	2,36,250		
Total	13,56,250	Total	13,56,250

Administration Overheads A/c

Particulars	Amount (₹)	Particulars	Amount (₹)
To General Ledger Adjustment A/c	6,25,000	By Finished Goods A/c	6,19,500
		By Overhead Adjustment A/c (Under-absorption)	5,500
Total	**6,25,000**	**Total**	**6,25,000**

Selling & Distribution Overheads A/c

Particulars	Amount (₹)	Particulars	Amount (₹)
To General Ledger Adjustment A/c	4,60,000	By Cost of Sales A/c	4,60,000
Total	**4,60,000**	**Total**	**4,60,000**

(iii) Reconciliation Statement

Particulars	Amount (₹)	Amount (₹)
Profits as per Cost Accounts		18,17,850
Add:		
– Production Overheads – over-absorbed	2,36,250	
– Dividend received	5,25,000	
– Interest from bank deposits	1,20,000	
– Write-down of Finished Stock (3,90,000 – 3,46,100)	43,900	
Total Additions		**9,25,150**
Less:		
– Administration Overheads – under-absorbed	5,500	
– Preliminary expenses written off	30,000	
– Goodwill written off	55,000	
– Fines	10,000	
– Interest on term loan	18,000	
– Loss on sale of equipment	20,000	
– Taxation	2,25,000	
Total Deductions		**3,63,500**
Profit as per Financial Accounts		**23,79,500**

13. Production Statement

For the Year Ended 31st March 2025

Particulars	Amount (₹)
Direct Materials	12,50,000
Direct Wages	9,80,000
Prime Cost	**22,30,000**

Factory Overheads	5,75,000
Cost of Production (COP)	**28,05,000**
Administration Overheads	5,40,000
Selling & Distribution Overheads	6,30,000
Cost of Sales	**39,75,000**
Profit	8,25,000
Sales Value	**48,00,000**

Calculation of Rates (2025-26)

1. **Percentage of Administration Overheads to Cost of Production (COP)** $= \frac{₹\,5,40,000}{₹\,28,05,000} \times 100 = 19.251\%$

2. **Percentage of Profit to cost = 8,25,000 / 39,75,000 = 20.755%**

3. **Selling & Distribution Overheads (20% Increase, Variable 65%)**

   new S&D = 1.20 × 6,30,000 = **₹ 7,56,000**

   S&D rate (revised) = 7,56,000 / 28,05,000 = **26.952%**

   Note: variable portion changed to 65% - that affects fixed/variable split but not the total used for recovery here

4. **Factory Overheads decreased by 10%:**

   new Factory OH = 0.90 × 5,75,000 = **₹5,17,500**

Factory OH rate (revised) = 5,17,500 / 9,80,000 = **52.806%**

Job Cost Sheet

Particulars	**Amount (₹)**
Direct Materials	3,20,000
Direct Wages	2,10,000
Prime Cost	**5,30,000**
Add: Factory Overhead applied (52.806% of DL)	1,10,893
Cost of Production	**6,40,893**

Add: Administration overhead (19.251% of CP)	1,23,378
Cost of Sales (CP + Admin)	**7,64,271**
Add: Selling & Distribution overhead (26.952% of CP)	1,72,733
Total Cost (before profit)	9,37,004
Add: Profit (20.755% on cost)	1,94,475
Sales Value (Quoted Price)	**11,31,479**

14. Statement of Equivalent Units (Process- I)

Input (Units)	Particulars	Output (Units)	Equivalent Production			
			Materials		Labour and Overheads	
			Units	(%)	Units	(%)
40,000	Introduced and completed	36,000	36,000	100	36,000	100
	Normal loss	2,000	-	-	-	-
	Closing stock	2,000	2,000	100	1,000	50
40,000		40,000	38,000		37,000	

Computation of cost per Equivalent Unit for each element of cost (Process- I)

Elements of Cost	Total Cost (₹)	Equivalent units	Cost per Equivalent units (₹)
Direct Materials	6,00,000	38,000	15.7895
Labour	1,20,000	37,000	3.2432
Factory Overheads	2,40,000	37,000	6.4865

Statement of Apportionment of Cost

Items	Elements	Equivalent units	Cost per unit (₹)	Cost (₹)	Total (₹)
Units introduced and completed	Materials	36,000	15.7895	5,68,422.00	
	Labour	36,000	3.2432	1,16,755.20	
	Overheads	36,000	6.4865	2,33,514.00	9,18,691.20
Closing stock	Materials	2,000	15.7895	31,579.00	
	Labour	1,000	3.2432	3,243.20	
	Overheads	1,000	6.4865	6,486.50	41,308.70

Process- I Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Materials	40,000	6,00,000	By Normal loss	2,000	-
To Labour		1,20,000	By Process II	36,000	9,18,691
To Overheads		2,40,000	By Closing stock	2,000	41,309
	40,000	9,60,000		40,000	9,60,000

Statement of Equivalent Units (Process- II)

Input (Units)	Particulars	Output (Units)	Equivalent Production			
			Materials		Labour and Overheads	
			Units	(%)	Units	(%)
36,000	Units transferred from Process- I					
	Normal loss	1,500	-	-	-	-
	Completed	32,000	32,000	100	32,000	100
	Closing stock (balancing figure)	2,500	2,500	100	1,250	50
36,000		36,000	34,500		33,250	

Computation of cost per Equivalent Unit for each element of cost (Process- I)

Elements of Cost	Total Cost (₹)	Equivalent units	Cost per Equivalent units (₹)
Cost of 36,000 units transferred from Process- I	9,18,691	34,500	26.6287
Labour	1,60,000	33,250	4.8120
Factory Overheads	2,00,000	33,250	6.0150

Statement of Apportionment of Cost

Items	Elements	Equivalent units	Cost per unit (₹)	Cost (₹)	Total (₹)
Units introduced and completed	Materials	32,000	26.6287	8,52,118.40	
	Labour	32,000	4.8120	1,53,984.00	
	Overheads	32,000	6.0150	1,92,480.00	11,98,582.40
Closing stock	Materials	2,500	26.6287	66,571.75	
	Labour	1,250	4.8120	6,015.00	
	Overheads	1,250	6.0150	7,518.75	80,105.50

Process- II Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Units introduced	36,000	9,18,691	By Normal loss	1,500	-
To Labour		1,60,000	By Finished stock	32,000	11,98,582
To Overheads		2,00,000	By Closing stock	2,500	80,109*
	36,000	12,78,691		36,000	12,78,691

*Difference arose due to rounding-off has been adjusted.

15. (i) Statement showing the apportionment of joint costs to P, Q and C

Products	P	Q	C	Total
Output (kg)	18,000	10,000	54,000	
Sales value at the point of split off (₹)	9,00,000 (₹ 50 x 18,000)	4,00,000 (₹ 40 x 10,000)	5,40,000 (₹ 10 x 54,000)	18,40,000
Joint cost apportionment on the basis of sales value at the point of split off (₹)	6,30,000 $\left(\frac{₹12,88,000}{₹18,40,000} x ₹9,00,000\right)$	2,80,000 $\left(\frac{₹12,88,000}{₹18,40,000} x ₹4,00,000\right)$	3,78,000 $\left(\frac{₹12,88,000}{₹18,40,000} x ₹5,40,000\right)$	12,88,000

(ii) Statement showing the cost per kg. of each product (indicating joint cost; further processing cost and total cost separately)

Products	P	Q	C
Joint costs apportioned (₹): (I)	6,30,000	2,80,000	3,78,000
Production (kg): (II)	18,000	10,000	54,000
Joint cost per kg (₹): (I ÷ II)	35	28	7
Further processing Cost per kg. (₹)	10 $\left(\frac{₹1,80,000}{18,000\text{kg}}\right)$	15 $\left(\frac{₹1,50,000}{10,000\text{kg}}\right)$	2 $\left(\frac{₹1,08,000}{54,000\text{kg}}\right)$
Total cost per kg (₹)	45	43	9

(iii) Statement showing the product wise and total profit for the period

Products	P	Q	C	Total
Sales value (₹)	12,24,000	2,50,000	7,92,000	
Add: Closing stock value (₹) (Refer to Working note 2)	45,000	2,15,000	90,000	
Value of production (₹)	12,69,000	4,65,000	8,82,000	26,16,000
Apportionment of joint cost (₹)	6,30,000	2,80,000	3,78,000	
Add: Further processing cost (₹)	1,80,000	1,50,000	1,08,000	
Total cost (₹)	8,10,000	4,30,000	4,86,000	17,26,000
Profit (₹)	4,59,000	35,000	3,96,000	8,90,000

Working Notes

1.

Products	P	Q	C
Sales value (₹)	12,24,000	2,50,000	7,92,000

Quantity sold (Kgs.)	17,000	5,000	44,000
Selling price ₹/kg	72 $\left(\frac{₹ 12,24,000}{17,000 kg}\right)$	50 $\left(\frac{₹ 2,50,000}{5,000 kg}\right)$	18 $\left(\frac{₹ 7,92,000}{44,000 kg}\right)$

2. Valuation of closing stock:

Since the selling price per kg of products P, Q and C is more than their total costs, therefore closing stock will be valued at cost.

Products	P	Q	C	Total
Closing stock (kgs.)	1,000	5,000	10,000	
Cost per kg (₹)	45	43	9	
Closing stock value (₹)	45,000 (₹ 45 x 1,000 kg)	2,15,000 (₹ 43 x 5,000 kg)	90,000 (₹ 9 x 10,000 kg)	3,50,000

(iv) Calculations for processing decision

Products	P	Q	C
Selling price per kg at the point of split off (₹)	50	40	10
Selling price per kg after further processing (₹) (Refer to working Note 1)	72	50	18
Incremental selling price per kg (₹)	22	10	8
Less: Further processing cost per kg (₹)	(10)	(15)	(2)
Incremental profit (loss) per kg (₹)	12	(5)	6

Product P and C has an incremental profit per unit after further processing, hence, these two products may be further processed.

However, further processing of product Q is not profitable hence, product Q shall be sold at split off point.

16. Working for Finding – Missing Figures

Cost Variance $_A$ = 0

Cost Variance $_{(A+B)}$ = ₹ 1,300 (A)

Yield Variance $_{(A+B)}$ = ₹ 270 (A)

Standard Cost and Actual Cost (Incomplete Information)

Raw Material	Standard Data			Actual Data		
	Qty. (Kg.) [SQ]	Price (₹) [SP]	Amount (₹) [SQ x SP]	Qty. (Kg.) [AQ]	Price (₹) [AP]	Amount (₹) [AQ x AP]
A	? ??	24	? ? ?	? ? ?	30	? ? ?
B	? ??	30	? ? ?	70	? ? ?	? ? ?
Total	? ? ?		? ??	? ??		? ? ?

Material Cost Variance $_A$ = Standard Cost – Actual Cost

⇒ 0 = (SQ_A × ₹ 24 – AQ_A × ₹ 30)

⇒ SQ_A = 1.25 AQ_A

Material Yield Variance $_{(A+B)}$ = *Average* Standard Price *per unit* of Standard Mix × [Total Standard Quantity (units) – Total Actual Quantity (units)]

⇒ ₹ 270 (A) = $\left(\dfrac{₹\,24 \text{ x } SQ_A + ₹\,30 \text{ x } SQ_B}{SQ_A + SQ_B}\right)$ × [(SQ_A + SQ_B) – (AQ_A+70)]

SQ_A = *SQ_B as Standard Mix is in ratio 1:1*

⇒₹ 270 (A) = $\left(\dfrac{₹\,24 \text{ x } SQ_A + ₹\,30 \text{ x } SQ_A}{SQ_A + SQ_A}\right)$ × [(SQ_A + SQ_A) – (AQ_A+70)]

⇒₹ 270 (A) = 27 × [2 x SQ_A – (AQ_A+70)]

⇒₹ 270 (A)	=	27 × [2 x 1.25 AQ_A – (AQ_A+70)]
⇒AQ_A	=	40 Kg.
As SQ_A	=	1.25 AQ_A
	=	1.25 × 40 Kg.
	=	50 Kg.
As SQ_B	=	SQ_A
	=	50 Kg.
Cost Variance $_{(A+B)}$	=	Standard Cost – Actual Cost
⇒ 1,300 (A)	=	(50 Kg. × ₹ 24 + 50 Kg. × ₹ 30) – (40Kg. ×₹ 30 +70 Kg. × AP_A)
⇒ AP_A	=	₹ 40

Standard Cost and Actual Cost (Complete Information)

Raw Material	Standard Data			Actual Data			Std. Cost of Actual Qty. (₹) [AQ x SP]
	Qty. (Kg.) [SQ]	Price (₹) [SP]	Amount (₹) [SQ x SP]	Qty. (Kg.) [AQ]	Price (₹) [AP]	Amount (₹) [AQ x AP]	
A	50	24	1,200	40	30	1,200	960
B	50	30	1,500	70	40	2,800	2,100
Total	100		2,700	110		4,000	3,060

Computation of Variances

Material Cost Variance	=	Standard Cost – Actual Cost
	=	SQ × SP – AQ × AP
(A)	=	₹1,200 – ₹1,200
	=	₹0
(B)	=	₹1,500 – ₹2,800
	=	₹1,300 (A)
Total	=	₹0 + ₹1,300 (A)
	=	₹1,300 (A)

Material Price Variance	=	Standard Cost of Actual Quantity – Actual Cost
	=	AQ × SP – AQ × AP
		Or
	=	AQ × (SP – AP)
(A)	=	40 Kg. × (₹24.00 – ₹30.00)
	=	₹240 (A)
(B)	=	70 Kg. × (₹30.00 – ₹40.00)
	=	₹700 (A)
Total	=	₹240 (A) + ₹700 (A)
	=	₹940 (A)
Material Usage Variance	=	Standard Cost of Standard Quantity for Actual Output – Standard Cost of Actual Quantity
	=	SQ × SP – AQ × SP
		Or
	=	SP × (SQ – AQ)
(A)	=	₹24 × (50 Kg. – 40 Kg.)
	=	₹240 (F)
(B)	=	₹30 × (50 Kg. – 70 Kg.)
	=	₹600 (A)
Total	=	₹240 (F) + ₹600 (A)
	=	₹360 (A)
Material Mix Variance	=	Total Actual Quantity (units) × (*Average* Standard Price *Per unit* of Standard Mix
		- Average Standard Price *per unit* of Actual Mix)

$$= 110\text{ Kg.} \times \left(\frac{₹\ 2,700}{100\text{ Kg.}} - \frac{₹\ 3,060}{110\text{ Kg.}}\right)$$

= ₹90 (A)

Material Yield Variance = *Average* Standard Price *per unit* of Standard Mix × [Total Standard Quantity (units) – Total Actual Quantity (units)]

$$= \left(\frac{₹\ 2,700}{100\text{ Kg.}}\right) \times (100\text{ Kg.} - 110\text{ Kg.})$$

= ₹ 270 (A)

Standard Output = Standard Input – Standard Loss

= 100 Kg. – 10 Kg.

= 90Kg.

Actual Output = 90 Kg.

(Actual Output and Standard Output are always equal numerically in any Material Variance Analysis)

17. Workings

(1)	All Overheads for one carton or 24*cans*	₹ 27
	Therefore, per *can* Overheads (₹27/24)	1.125
	Fixed Overheads Allocated for 1,50,000 *cans*:	₹ 1,12,500
	Per *can* Fixed Overheads (₹1,12,500 / 1,50,000*cans)*	₹ 0.75
	Variable Overheads per *can* (₹1.125 – ₹0.75)	₹ 0.375
(2)	Direct Wage per carton	₹ 36
	Per *can* (₹36/24)	₹ 1.50
(3)	Direct Materials per carton	₹ 54
	Per *can* (₹54/24)	₹ 2.25

(4) Cost of making one empty *can*:

	Cost per can of 'EXE' (₹)	Cost % empty *can*	Cost empty *can* (₹)	Cost of per can of 'EXE' without empty can (₹)
Direct Material	2.250	20	0.4500	1.8000
Direct Wages	1.500	10	0.1500	1.3500
Variable Overheads	0.375	10	0.0375	0.3375
Total	4.125		0.6375	3.4875

(5) Cost of manufacturing/buying of 1,50,000 empty *cans* of 'EXE':

	Empty can Cost (₹)	If empty can made (₹)	If empty can purchased (₹)
Direct Material	0.4500	67,500.00	-----
Direct Wages	0.1500	22,500.00	-----
Variable Overheads	0.0375	5,625.00	-----
Purchase Price	0.6750	-----	1,01,250.00
Total		95,625.00	1,01,250.00

The company should manufacture the empty *cans* for a production volume of 1,50,000 'EXE' *cans* as capacity is available and cost of manufacture is lower.

(6) After the level of 1,50,000 empty cans, the company has to install a new machine involving a total additional Fixed Overheads of ₹ 7,500. The cost of making and buying the additional *cans* of 25,000 and 75,000 will be as follows:

	Cost per *can* (₹)	Make (₹)	Buy (₹)	Make (₹)	Buy (₹)
		25,000 *cans*		75,000 *cans*	
Direct Material	0.4500	11,250.00	-	33,750.00	-
Direct Wages	0.1500	3,750.00	-	11,250.00	-

Variable Overheads	0.0375	937.50	-	2,812.50	-
Additional Overheads		7,500.00	-	7,500.00	-
Purchase Price	0.6750	-	16,875.00	-	50,625.00
Total		23,437.50	16,875.00	55,312.50	50,625.00

The cost of buying additional empty containers at both the levels is lower than the cost of their manufacture.

(a) If the company increases production to 1,75,000 cans of 'EXE', 1,50,000 empty cans should be manufactured and additional 25,000 cans should be purchased at ₹16,875 [Refer W.N. 5&6]

If the company increases production to 2,25,000 cans of 'EXE', 1,50,000 empty cans should be manufactured and additional 75,000 cans should be purchased at a cost of ₹ 50,625. [Refer W.N. 5&6]

(b) Additional fixed overheads to be incurred on a new machine: ₹7,500 Savings per unit if empty cans are made instead of buying:

₹ 0.675 – ₹ 0.6375 = ₹ 0.0375

Minimum additional quantity of empty containers to be made to recover the additional fixed costs:

₹ 7,500/ ₹0.0375 = 2,00,000 empty cans

Installation of the new machine for the manufacture of empty cans will be economical at production level of 3,50,000 cans per month.

(c) Evaluation of the profitability on sale of "EXE" at the three levels.

	Per *can* (₹)	**1,50,000 *can* (₹)**	**1,75,000 *can* (₹)**	**2,25,000 *can* (₹)**
Sales	5.0000	7,50,000.00	8,75,000.00	11,25,000.00
Less: Direct Material	1.8000	2,70,000.00	3,15,000.00	4,05,000.00
Direct Wages	1.3500	2,02,500.00	2,36,250.00	3,03,750.00
Variable Overheads	0.3375	50,625.00	59,062.50	75,937.50

Empty *can* made	0.6375	95,625.00	95,625.00	95,625.00
Empty *can* purchases	0.6750		16,875.00	50,625.00
Net Gain		1,31,250.00	1,52,187.50	1,94,062.50

18. Material Purchase Budget

(in quantities and value)

Particulars	Total
Material Consumption (Kg.) A (1,740 units x 5 Kg. = 8,700) B (2,000 units x 3 Kg. = 6,000)	14,700
Add: Closing Balance of Material (Kg.) (W.N. 3)	2,450
Less: Anticipated Opening Balance of Material (Kg.)	2,150
Total Quantity of Material (kg.) to be purchased	15,000
Total Value of Material to be purchased (₹) (15,000 Kg. x ₹12)	₹ 1,80,000

Direct Workers Wages Budget

(showing hours required and wages paid)

Particulars	Total
Standard Hours for Budgeted Production A (1,740 units x 5 hrs.= 8,700) B (2,000 units x 4 hrs.= 8,000)	16,700
Standard Hours for Budgeted Production at Targeted Efficiency Ratio (W.N. 4)	20,875
Add: Non Productive Downtime (20% x 20,875 hours)	4,175
Total Labour Hours Required	25,050
Less: Normal Labour Hours (45 workers x 12 weeks x 5 days x 8 hours)	21,600
Difference (Overtime hours)	3,450

Wages for normal hours (21,600 hours x ₹ 8)	₹1,72,800
Overtime Wages (3,450 x ₹12)	₹41,400
Total Wages	₹2,14,200

Working Notes:

1. Computation of Closing Stock Balance of Products A and B

   Budgeted Period of Sales (In days) = 12 weeks x 5 days = 60 days

   Closing Stock of Product A (Units) (15 days sales)

   $$= \frac{1,800 \text{ units} \times 15 \text{ days}}{60 \text{ days}} = 450 \text{ units}$$

   Closing stock of Product B (units) (20 days sales)

   $$= \frac{2,400 \text{ units} \times 20 \text{ days}}{60 \text{ days}} = 800 \text{ units}$$

2. Production Budget (Units)

	Products	
	A	**B**
Sales in units (60 days)	1,800	2,400
Add: Closing Stock (W.N. 1)	450	800
	2,250	3,200
Less: Anticipated Opening Balance	510	1,200
Total Number of Units to be produced	1,740	2,000

3. Closing Balance of Material (Kg.)

   Closing Balance of Material (10 days consumption)

   $$= \frac{14,700 \text{ kg}}{60 \text{ days}} \times 10 \text{ days} = 2,450 \text{ Kg.}$$

4. Standard Hours for Budgeted Production at targeted 80% efficiency ratio $= \frac{16,700 \text{ Hrs.}}{80} \times 100 = 20,875 \text{ Hrs.}$

19. (a) Type of costs categorised based on its use in Managerial Decision Making

(i) **Pre-determined Cost -** A cost which is computed in advance before production or operations start, on the basis of specification of all the factors affecting cost, is known as a pre-determined cost.

(ii) **Standard Cost -** A pre-determined cost, which is calculated from managements 'expected standard of efficient operation' and the relevant necessary expenditure. It may be used as a basis for price fixation and for cost control through variance analysis.

(iii) **Marginal Cost -** The amount at any given volume of output by which aggregate costs increases if the volume of output is increased or decreased by one unit.

(iv) **Estimated Cost -** Kohler defines estimated cost as "the expected cost of manufacture, or acquisition, often in terms of a unit of product computed on the basis of information available in advance of actual production or purchase". Estimated costs are prospective costs since they refer to prediction of costs.

(v) **Differential Cost -** (Incremental and decremental costs). It represents the change (increase or decrease) in total cost (variable as well as fixed) due to change in activity level, technology, process or method of production, etc. For example, if any change is proposed in the existing level or in the existing method of production, the increase or decrease in total cost or in specific elements of cost as a result of this decision will be known as incremental cost or decremental cost.

(vi) **Imputed Costs -** These costs are notional costs which do not involve any cash outlay. Interest on capital, the payment for which is not actually made, is an example of imputed cost. These costs are similar to opportunity costs.

(vii) **Capitalized Costs -** These are costs which are initially recorded as assets and subsequently treated as expenses.

Example, installation expenses on the erection of a machine are added to the cost of a machine.

(viii) Product Costs - These are the costs which are associated with the purchase and sale of goods (in the case of merchandise inventory). In the production scenario, such costs are associated with the acquisition and conversion of materials and all other manufacturing inputs into finished product for sale. Hence, under marginal costing, variable manufacturing costs and under absorption costing, total manufacturing costs (variable and fixed) constitute inventoriable or product costs.

(ix) Opportunity Cost - This cost refers to the value of sacrifice made or benefit of opportunity foregone in accepting an alternative course of action. For example, a firm financing its expansion plan by withdrawing money from its bank deposits. In such a case the loss of interest on the bank deposit is the opportunity cost for carrying out the expansion plan.

(x) Out-of-pocket Cost - It is that portion of total cost, which involves cash outflow. This cost concept is a short-run concept and is used in decisions relating to fixation of selling price in recession, make or buy, etc. Out–of–pocket costs can be avoided or saved if a particular proposal under consideration is not accepted.

(xi) Shut down Costs - Those costs, which continue to be, incurred even when a plant is temporarily shut-down e.g. rent, rates, depreciation, etc. These costs cannot be eliminated with the closure of the plant. In other words, all fixed costs, which cannot be avoided during the temporary closure of a plant, will be known as shut down costs.

(xii) Sunk Costs - Historical costs incurred in the past are known as sunk costs. They play no role in decision making in the current period. For example, in the case of a decision relating to the replacement of a machine, the written down value of the existing machine is a sunk cost and therefore, not considered.

(xiii) Absolute Cost - These costs refer to the cost of any product, process or unit in its totality. When costs are presented in a statement form, various cost components may be shown in absolute amount or as a percentage of total cost or as per unit cost or all together. Here the costs depicted in absolute amount may be called absolute costs and are base costs on which further analysis and decisions are made.

(xiv) Discretionary Costs - Such costs are not tied to a clear cause and effect relationship between inputs and outputs. They usually arise from periodic decisions regarding the maximum outlay to be incurred. Examples include advertising, public relations, executive training etc.

(xv) Period Costs - These are the costs, which are not assigned to the products but are charged as expenses against the revenue of the period in which they are incurred. All non-manufacturing costs such as general & administrative expenses, selling and distribution expenses are recognised as period costs.

(xvi) Engineered Costs - These are costs that result specifically from a clear cause and effect relationship between inputs and outputs. The relationship is usually personally observable. Examples of inputs are direct material costs, direct labour costs etc. Examples of output are cars, computers etc.

(xvii) Explicit Costs - These costs are also known as out-of-pocket costs and refer to costs involving immediate payment of cash. salaries, wages, postage and telegram, printing and stationery, interest on loan etc. are some examples of explicit costs involving immediate cash payment.

(xviii) Implicit Costs - These costs do not involve any immediate cash payment. They are not recorded in the books of account. They are also known as economic costs.

(b)

Cost Pool	**Cost Driver**
Quality Control	• Number of Inspections • Product Units Produced
Research and Development	• Number of research projects • Personnel hours on a project
Machine Maintenance	• Machine Hours
Employee Training Costs	• Number of Training Hours • Employees Trained
Customer Service	• Number of service calls • Number of products serviced • Hours spent on servicing products

(c) Limitations of Budgetary Control System

Points	**Description**
1. Based on Estimates	Budgets are based on a series of estimates, which are based on the conditions prevalent or expected at the time budget is established. It requires revision in plan if conditions change.
2. Time factor	Budgets cannot be executed automatically. Some preliminary steps are required to be accomplished before budgets are implemented. It requires proper attention and time of management. Management must not expect too much during the initial development period.
3. Co-operation Required	Staff co-operation is usually not available during the initial budgetary

	control exercise. In a decentralised organisation, each unit has its own objective and these units enjoy some degree of discretion. In this type of organisation structure, coordination among different units is required. The success of the budgetary control depends upon willing co-operation and teamwork.
4. Expensive	The implementation of budget is somewhat expensive. For successful implementation of the budgetary control, proper organisation structure with responsibility is prerequisite. Budgeting process start from the collection of information to for preparing the budget and performance analysis. It consumes valuable resources (in terms of qualified manpower, equipment, etc.) for this purpose; hence, it is an expensive process.
5. Not a substitute for management	Budget is only a managerial tool and must be intelligently applied for management to get benefited. Budgets are not a substitute for good management.
6. Rigid document	Budgets are sometime considered as rigid documents. But in reality, an organisation is exposed to various uncertain internal and external factors. Budget should be flexible enough to incorporate ongoing developments in the internal and external factors affecting the very purpose of the budget.

(d)

Cost Control	Cost Reduction
1. Cost control aims at maintaining the costs in accordance with the established standards.	1. Cost reduction is concerned with reducing costs. It challenges all standards and endeavours to improvise them continuously
2. Cost control seeks to attain lowest possible cost under existing conditions.	2. Cost reduction recognises no condition as permanent, since a change will result in lower cost.
3. Cost control is a preventive function.	3. Cost reduction is a corrective function. It operates even when an efficient cost control system exists.
4. Cost control ends when targets are achieved.	4. Cost reduction has no visible end and is a continuous process.

PAPER – 5: AUDITING AND ETHICS

QUESTIONS

PART – I: Multiple Choice Questions based on Case Scenarios

Case Scenario

CA Anuj is in practice for the past few years as proprietor of Anuj & Associates. He is very keen on documentation aspects while performing assurance engagements. He handles audits of various corporate and non-corporate clients with team members. The proprietorship concern also handles review engagements as well as engagements to report on prospective financial information as per needs and requirements of its clientele. Presently, it is handling audit of "Buddy Enterprises", a partnership firm, for year 2024-25 in middle of busy audit season. This business enterprise is also availing cash credit limit of ₹ 10 lacs from CDB Bank. The firm's partnership deed contains a clause stipulating audit of its financial statements by a firm of Chartered Accountants every year. However, there is no legal requirement to get financial statements of partnership firm audited. Its Profit before tax is ₹ 15 lacs.

The audit team has planned everything right from risk assessment procedures to tests of controls as well as substantive procedures. Audit plan and audit programme have been duly documented. However, during the course of audit, some information has come to notice of audit team which requires change in audit plan and could lead to revision of audit strategy. J, junior team member is pondering over documentation requirements in this peculiar situation.

Another team member, K has understood meaning of "sufficient appropriate" evidence while conducting audits. However, he is wondering over obtaining sufficient appropriate evidence in case of assurance engagements which may not be in nature of audit. At the initial stage, audit team has determined materiality for financial statements as whole @ 5% of Profit before tax (i.e. 5% of ₹ 15 lacs = ₹ 75000). K is also concerned about documenting materiality as part of audit documentation.

K vividly remembers that an assurance engagement involves three party relationship. However, another team member Q confronts him with meaning of three-party relationship.

In accordance with terms of sanction letter, it has been stipulated by CDB bank that partnership firm is required to submit quarterly stock statements to bankers. During quarter ended 31st December 2024, value of stocks declared by firm to bankers is ₹ 10 lacs and margin stipulated in sanction letter is 20%.

Based upon above, answer the following questions: -

1. Considering dilemma of J described in case scenario, which of the following statements is most appropriate?

   (a) Only overall audit strategy needs to be documented by audit team.

   (b) Only audit plan needs to be documented by audit team.

   (c) Both overall audit strategy and audit plan need to be documented by audit team.

   (d) Overall audit strategy, audit plan and any significant changes made during the audit engagement to the overall audit strategy or the audit plan and the reasons for such changes need to be documented by audit team.

2. Guide K regarding most appropriate statement concerning obtaining of "sufficient appropriate" evidence as described in case scenario.

   (a) Sufficient appropriate evidence is to be obtained only in case of statutory audits. In the given case, there is no requirement of law to get accounts audited. Hence, there is no need to obtain sufficient appropriate evidence.

(b) Sufficient appropriate evidence is to be obtained in audit engagements, review engagements and engagements involving examination of prospective financial information.

(c) Sufficient appropriate evidence is to be obtained in audit engagements and review engagements.

(d) Sufficient appropriate evidence is to be obtained in all audit engagements.

3. Advise K regarding most appropriate statement in respect of documenting materiality.

(a) Materiality of ₹ 75,000 needs to be documented.

(b) Performance materiality which is set at more than materiality for financial statements as a whole requires to be documented along with materiality of ₹ 75,000.

(c) Performance materiality which is set at less than materiality for financial statements as a whole requires to be documented along with materiality of ₹ 75,000.

(d) Performance materiality (for certain transactions) which is set at par level materiality for financial statements as a whole requires to be documented along with materiality of ₹ 75,000.

4. Which of the following combinations best describes three party relationship in context of audit of partnership firm?

(a) Anuj & Associates, management/partners of Buddy enterprises and income tax authorities.

(b) Individual team members assisting CA Anuj, management/partners of Buddy enterprises and income tax authorities.

(c) Anuj & Associates, management/partners of Buddy enterprises and bankers of Buddy Enterprises.

(d) Anuj & Associates, management/partners of Buddy enterprises and intended users.

5. Which of the following statements is true regarding drawing power on basis of facts given in case scenario?

   (a) DP is ₹ 10 lacs and firm can withdraw funds up to ₹ 10 lacs.

   (b) DP is ₹ 12 lacs and firm can withdraw funds up to ₹ 12 lacs.

   (c) DP is ₹ 8 lacs and firm can withdraw funds up to ₹ 10 lacs.

   (d) DP is ₹ 8 lacs and firm can withdraw funds up to ₹ 8 lacs.

General MCQs

6. The auditor of PAC Pvt Ltd., a pickle manufacturing company, notices that the company is expanding its operations into North India States in Financial Year 2024-25, but the related internal control system remains the same to handle the increased volume of transactions and consequently auditor concludes that material misstatements would not be prevented, or detected and corrected by the entity's internal control. Type of risk involved in this situation is:

   (a) Detection Risk.

   (b) Control Risk.

   (c) Inherent Risk.

   (d) Audit Risk.

7. The auditor of XYZ Ltd. performed audit procedures to obtain sufficient appropriate audit evidence covering each and every area impacting the financial statements of the company by 20th July 2025. Before signing the auditor's report on 16th September 2025, the auditor became aware of a fire accident at the company's Yarn Factory on 10th August 2025, destroying a large part of inventory. Auditor's responsibilities in such a situation are:

   (i) Perform audit procedures designed to obtain sufficient appropriate audit evidence that all events occurring between the date of financial statements and the date of auditor's report that require adjustment or disclosure in the financial statements have been identified.

   (ii) If such subsequent events that require adjustment or disclosure in the financial statements have been identified, it shall be determined whether each such event is appropriately reflected in those financial statements.

(iii) Obtain a written representation from management that all events occurring subsequent to the date of financial statements and for which appropriate adjustment has been done or required disclosure has been made.

(a) (i)

(b) (i) and (ii)

(c) (i) and (iii)

(d) (i), (ii) and (iii)

8. While conducting audit of Sandhya Ltd., having turnover of ₹ 200 crores and borrowings from Banks ₹ 50 crores. Audit team is determining that the accounting policies and methods of recording debt are appropriate and applied consistently and also verifying loan balances with reference to sanction letter/ loan agreement. By doing above, audit team is trying to satisfy about:

(a) Completeness.

(b) Existence.

(c) Valuation.

(d) Disclosure.

PART II – Descriptive Questions

Chapter 1 - Nature, Objective and Scope of Audit

9. Micky & Associates, Chartered Accountants, were appointed as the auditors of STI Ltd., a Textile manufacturing company, for the financial year 2024-25. Management of the company discussed with the newly appointed auditors that the company has been suffering losses for the last few years and also the Directors are suspecting that accounts are not properly maintained and might contain errors and frauds. So, in the given situation, management of the company requested the auditors to give complete assurance/guarantee about the financial statements of the company.

Can the auditors give complete assurance/guarantee in the above situation? Discuss with reference to objectives of Audit.

Chapter 2 - Audit Strategy, Audit Planning and Audit Programme

10. MICA Associates., Chartered Accountants, are appointed as statutory auditors of B Ltd. for the first time in F.Y. 2024-25. Before commencing the audit, the engagement partner considers the integrity of Directors, evaluates compliance with ethical requirements including independence, sends an engagement letter to avoid misunderstanding.

    Explain the above stage of the audit planning process as considered by the engagement partner.

Chapter 3 - Risk Assessment and Internal Control

11. SPOM & Co. is a firm of Chartered Accountants established in 2010 with major focus on Audit Practice. In F.Y. 2024-25, firm has four partners and 15 articled assistants handling various Internal audit and Statutory Audit assignments. Partners and articled assistants meet monthly to discuss various aspects of practical as well as academic issues. While discussing about various aspects of Audit Risk, Risk of material misstatement etc., Partner S asks one of the articled assistants to explain the concept of misstatement with the help of at least five examples.

Chapter 4 - Audit Evidence

12. With respect to SA 505, "External Confirmations", name the appropriate term that may be used in relation to the below statements:

    (a) Audit evidence obtained as a direct written response to the auditor from a third party (the confirming party), in paper form, or by electronic or other medium.

    (b) A request that the confirming party respond directly to the auditor indicating whether the confirming party agrees or disagrees with the information in the request, or providing the requested information.

    (c) A request that the confirming party respond directly to the auditor only if the confirming party disagrees with the information provided in the request.

(d) A failure of the confirming party to respond, or fully respond, to a positive confirmation request, or a confirmation request returned undelivered.

(e) A response that indicates a difference between information requested to be confirmed, or contained in the entity's records, and information provided by the confirming party.

13. SIME & Co., Chartered Accountants, have been appointed as statutory auditors of Delight & Delight, a Partnership firm, for the first time for F.Y. 2024-25. The previous year's financial statements were not subject to audit. During the audit, the new auditors noticed differences in some opening balances brought forward from previous year. With respect to relevant SA, explain the objective of Auditor with respect to opening balances.

Chapter 5 - Audit of Items of Financial Statements

14. In the Balance Sheet of Watches Ltd., additions worth ₹ 20 Crores are appearing under Property, Plant & Equipment. Auditor of the company, in addition to the procedures undertaken for verifying completeness of additions to PPE during the period under audit, while performing testing of additions also want to verify that all PPE purchase invoices are in the name of the company. Name the relevant assertion for verification of ownership and procedure to verify the same.

Chapter 6 - Audit Documentation

15. FSY & Co., Chartered Accountants, completed the statutory audit of HIMMAT Ltd. for the year ended 31st March 2025. The audit partner instructed the audit team to finalize and assemble the audit documentation immediately after signing the auditor's report on 15th September 2025. However, audit team took this instruction lightly and forgot to assemble the audit documentation. After six months, audit team started assembling the file and completed its documentation. When audit partner became aware of this, he called a meeting of staff and discussed policies and procedures for timely completion of audit files, changes to audit documentation and period of retention of engagement documentation.

Explain various points discussed by audit partner with reference to relevant standard on auditing.

Chapter 7 - Completion and Review

16. GEM Ltd has appointed GSP & Co., Chartered Accountants as its Statutory Auditors. While conducting its audit, the audit partner requested the management to provide a written representation that it has fulfilled its responsibility for the preparation of financial statements in accordance with the applicable financial reporting framework. The Manager (Finance) is of the view that since all records and evidence have already been given, a separate written representation is not required. Explain.

Chapter 8 - Audit Report

17. SA 701 deals with the auditor's responsibility to communicate key audit matters in the auditor's report. It is intended to address both the auditor's judgement as to what to communicate in the auditor's report and the form and content of such communication.

    In the above context, explain the purpose of communicating key audit matters.

Chapter 9 - Special Features of Audit of Different Type of Entities

18. You are appointed as an auditor of Great Eastern Hotel, a Five Star hotel, for Financial Year 2024-25. As an auditor what are the special points that need to be considered in verifying the Inventories in the nature of food and beverages?

Chapter 10 - Audit of Banks

19. While verifying provisions of advances of a branch of HFC Bank as part of statutory branch audit, CA Mini notices that credit facilities consisting of term loan to JT Mills have been classified under doubtful asset (D2) category.

    The outstanding balance in above term loan account as on 31.3.2025 is ₹ 150 lakhs. Value of security held in account is ₹ 100 lakhs. The branch has made provision of ₹ 75 lakhs. Is provision made by branch proper? Discuss.

Chapter 11 - Ethics and Terms of Audit Engagements

20. CA Jatin is auditor of XYZ Pharma, his brother-in-law's Partnership Firm. His family relationship creates a conflict of interest that could unduly influence his professional judgment, making it difficult for him to provide an unbiased opinion on the firm's financial statements.

    In another assignment of Tax Audit of a company, CA Jatin is also required to prepare the company's tax return. He fails to check the latest provision of the Income-tax Act which allows for a major new deduction resulting in the client overpaying income tax.

    CA Jatin is also preparing GST returns of one of his clients. The client instructs him to intentionally understate the sales figures to reduce the tax liability.

    Identify and explain the fundamental principles of professional ethics involved in above situations.

SUGGESTED ANSWERS

PART – I: Answers to Multiple Choice Questions

MCQ No.	Answer
1.	(d)
2.	(b)
3.	(c)
4.	(d)
5.	(d)
6.	(b)
7.	(d)
8.	(c)

PART – II: Answers to Descriptive Questions

9. As per SA 200 "Overall Objectives of the Independent Auditor and the Conduct of an Audit in Accordance with Standards on Auditing", auditor's objective is to obtain a reasonable assurance whether financial statements as a whole are free from material misstatement whether due to fraud or error.

(i) Reasonable assurance is to be distinguished from absolute assurance. Absolute assurance is a complete assurance or a guarantee that financial statements are free from material misstatements. However, reasonable assurance is not a complete guarantee. Although it is a high-level of assurance, but it is not complete assurance.

Audit of financial statements is carried out by the auditor with professional competence and skills in accordance with Standards on Auditing. Audit procedures are applied in accordance with SAs, audit evidence is obtained and evaluated. On basis of that, conclusions are drawn, and opinion is formed. It leads to high level of assurance which is called reasonable assurance but it is not absolute assurance.

(ii) Misstatements in financial statements can occur due to fraud or error or both. The auditor seeks to obtain reasonable assurance whether financial statements as a whole are free from material misstatements caused by fraud or error. He has to see effect of misstatements on financial statements as a whole, in totality.

(iii) Obtaining reasonable assurance that financial statements as a whole are free from material misstatements enables the auditor to express an opinion on whether the financial statements are prepared, in all material respects, in accordance with an applicable financial reporting framework.

(iv) The opinion is reported and communicated in accordance with audit findings through a written report as required by Standards on Auditing.

Thus, it can be concluded that auditors can provide reasonable assurance instead of absolute assurance.

10. Preliminary engagement activities

The auditor considers whether relationship with client should be continued and whether ethical requirements including independence continue to be complied with. It includes:

(A) Performing procedures regarding the continuance of the client relationship. It should be ensured that appropriate procedures regarding the acceptance and continuance of client relationships and audit engagements have been followed and that conclusions reached in this regard are appropriate.

The firm should obtain information considered necessary in the circumstances before accepting an engagement with a new client, when deciding whether to continue an existing engagement, and when considering acceptance of a new engagement with an existing client.

Matters such as integrity of principal owners and key management, competence of engagement team to perform the audit engagement and implications of matters that have arisen during current and previous audit engagement may need to be considered.

Besides, in case of initial engagements, communication with predecessor auditor should be made, where there has been a change of auditors.

(B) Evaluating compliance with ethical requirements, including independence. The auditor shall continuously evaluate compliance with ethical requirements including independence. "Independence" means that the judgement of a person is not subordinate to the wishes or direction of another person who might have engaged him.

Throughout the audit engagement, the engagement partner shall remain alert, through observation and making inquiries as necessary, for evidence of non-compliance with relevant ethical requirements by members of the engagement team.

(C) Establishing an understanding of terms of engagement. It is in the interests of both the entity and the auditor that the auditor sends an audit engagement letter before the commencement of the audit to help avoid misunderstandings with respect to the audit. It ensures that there is no confusion with the client regarding terms of the engagement.

11. Misstatement refers to difference between the amount, classification, presentation, or disclosure of a reported financial statement item and the amount, classification, presentation, or disclosure that is required for the item to be in accordance with the applicable financial reporting framework. Misstatements can arise from error or fraud.

Few examples of misstatements are:

i. Charging an item of capital expenditure to revenue or vice versa.

ii. Difference in disclosure of a financial statement item vis-vis its requirement in applicable financial reporting framework.

iii. Selection or application of inappropriate accounting policies.

iv. Difference in accounting estimate of a financial statement item vis-a-vis its appropriateness in applicable financial reporting framework.

v. Intentional booking of fake expenses in statement of profit and loss.

vi. Overstating receivables in financial statements by not writing off irrecoverable debts.

vii. Overstating or understating inventories.

12.

(a)	Audit evidence obtained as a direct written response to the auditor from a third party (the confirming party), in paper form, or by electronic or other medium.	External Confirmation.
(b)	A request that the confirming party respond directly to the auditor indicating whether the confirming party agrees or	Positive Confirmation request.

	disagrees with the information in the request, or providing the requested information.	
(c)	A request that the confirming party respond directly to the auditor only if the confirming party disagrees with the information provided in the request.	Negative confirmation request.
(d)	A failure of the confirming party to respond, or fully respond, to a positive confirmation request, or a confirmation request returned undelivered.	Non-response.
(e)	A response that indicates a difference between information requested to be confirmed, or contained in the entity's records, and information provided by the confirming party.	Exception.

13. As per SA 510, "Initial Audit Engagements – Opening Balances", in conducting an initial audit engagement, the objective of the auditor with respect to opening balances is to obtain sufficient appropriate audit evidence about whether:

(a) Opening balances contain misstatements that materially affect the current period's financial statements; and

(b) Appropriate accounting policies reflected in the opening balances have been consistently applied in the current period's financial statements, or changes thereto are properly accounted for and adequately presented and disclosed in accordance with the applicable financial reporting framework.

14. Relevant assertion for verification of ownership of PPE in the given situation is Rights and Obligation.

Procedure to verify the ownership rights over PPE is as under:

i. The auditor while performing testing of additions should also verify that all PPE purchase invoices are in the name of the entity that entitles legal title of ownership to the respective entity.

ii. For all additions to land and building in particular, the auditor should check the conveyance deed/ sale deed to verify whether the entity is the legal and valid owner or not.

iii. The auditor should insist and verify the original title deeds for all immoveable properties held as at the balance sheet date.

iv. In case the entity has given such immoveable property as security for any borrowings and the original title deeds are not available with the entity, the auditor should request the entity's management for obtaining a confirmation from the respective lenders that they are holding the original title deeds of immoveable property as security.

v. In addition, the auditor should also verify the register of charges, available with the entity to assess that any charge has been created against the PPE.

15. As per SA 230, "Audit Documentation", the auditor shall assemble the audit documentation in an audit file and complete the administrative process of assembling the final audit file on a timely basis after the date of the auditor's report.

SQC 1, "Quality Control for Firms that perform Audits and Review of Historical Financial Information, and other Assurance and related services", requires firms to establish policies and procedures for the timely completion of the assembly of audit files.

An appropriate time limit within which to complete the assembly of the final audit file is ordinarily not more than 60 days after the date of the auditor's report. The completion of the assembly of the final audit file after the date of the auditor's report is an administrative process that does not involve the performance of new audit procedures or the drawing of new conclusions.

Changes may, however, be made to the audit documentation during the final assembly process, if they are administrative in nature.

Examples of such changes include:

- ✓ Deleting or discarding superseded documentation.
- ✓ Sorting, collating and cross-referencing working papers.
- ✓ Signing off on completion checklists relating to the file assembly process.
- ✓ Documenting audit evidence that the auditor has obtained, discussed and agreed with the relevant members of the engagement team before the date of the auditor's report.

After the assembly of the final audit file has been completed, the auditor shall not delete or discard audit documentation of any nature before the end of its retention period.

SQC 1 requires firms to establish policies and procedures for the retention of engagement documentation. The retention period for audit engagements ordinarily is no shorter than seven years from the date of the auditor's report, or, if later, the date of the group auditor's report.

16. In the given case, GSP & Co., auditor of GEM Ltd. requested the management to provide written representation that it has fulfilled its responsibility for the preparation of financial statements in accordance with the applicable financial reporting framework. The Manager (Finance) is of the view that since all records and evidence have already been given, a separate written representation is not required. View of Manager (Finance) is not correct because written representations about management responsibilities are necessary due to the following reasons:

Audit evidence obtained during the audit that management has fulfilled its responsibilities regarding preparation of financial statements and about information provided and completeness of transactions is not sufficient without obtaining confirmation from management that it believes that it has fulfilled those responsibilities. This is because the auditor is not able to judge solely on other audit evidence whether management has prepared and presented the financial statements and provided information to the auditor on the basis of the agreed acknowledgement and understanding of its responsibilities.

For example, the auditor could not conclude that management has provided the auditor with all relevant information agreed in terms of the audit engagement without asking it whether, and receiving confirmation that, such information has been provided.

The written representations requiring fulfilment of management responsibilities in relation to above draw on the agreed acknowledgement and understanding of management of its responsibilities in the terms of the audit engagement by requesting confirmation that it has fulfilled them. The auditor may also ask management to reconfirm its acknowledgement and understanding of those responsibilities in written representations.

This is particularly appropriate when: -

- Those who signed the terms of the audit engagement on behalf of the entity no longer have the relevant responsibilities;
- The terms of the audit engagement were prepared in a previous year;
- There is any indication that management misunderstands those responsibilities; or
- Changes in circumstances make it appropriate to do so.

17. **Purpose of Communicating Key Audit Matters:** As per SA 701, "Communicating Key Audit Matters in the Auditor's Report", the purpose of communicating key audit matters is to enhance the communicative value of the auditor's report by providing greater transparency about the audit that was performed. Communicating key audit matters provides additional information to intended users of the financial statements to assist them in understanding those matters that, in the auditor's professional judgement, were of most significance in the audit of the financial statements of the current period. Communicating key audit matters may also assist intended users in understanding the entity and

areas of significant management judgement in the audited financial statements.

18. **Verification of inventories in the nature of food and beverages:** The inventories in any hotel are both readily portable and saleable particularly the food and beverage inventories. It is therefore extremely important that all movements and transfers of such inventories should be properly documented to enable control to be exercised over each individual stores' areas and sales point. The auditor should carry out tests to ensure that all such documentation is accurately processed. Therefore, following points may be noted in this regard:

(a) Areas where inventories are kept must be kept locked and the key retained by the departmental manager.

(b) The key should be released only to trusted personnel and unauthorized persons should not be permitted in the stores area.

(c) Many hotels use specialized professional valuers to count and value the inventories on a continuous basis throughout the year.

(d) The auditor should ensure that all inventories are valued at the year end and that he should himself be present at the year-end physical verification, to the extent practicable, having regard to materiality consideration and nature and location of inventories

19. The provision for advances falling in D2 category is required @ 40% of secured portion and 100% of unsecured portion.

Therefore, provision in above case is arrived as under: -

Outstanding balance as on 31.3.25	150 lakhs
Less: Value of security	100 lakhs
Unsecured portion	50 lakhs

Therefore, outstanding balance of ₹ 150 lakhs is secured to the extent of ₹ 100 lakhs and balance ₹ 50 lakhs is unsecured.

Provision calculation	₹
40% of secured part i.e., 40% of ₹ 100 lakhs	40 lakhs
100% of unsecured part i.e.,100% of ₹ 50 lakhs	50 lakhs
Required provision	90 lakhs

In view of above, provision made by branch is not proper. It should have made provision of ₹ 90 lakhs instead of ₹ 75 lakhs.

20. **(i)** In the given case, CA Jatin's family relationship creates a conflict of interest that could unduly influence his professional judgment. This indicates that "Objectivity" principle is being violated.

The principle of objectivity requires an auditor not to compromise professional judgment because of bias, conflict of interest or undue influence of others.

It requires that a professional accountant shall not undertake a professional activity if a circumstance or relationship unduly influences the accountant's professional judgment regarding that activity.

(ii) In the instant case, CA Jatin did not check the latest provision of the Income-tax Act while preparing return of income-tax, which allows for a major new deduction resulting in the client overpaying income tax. This indicates "Professional competence and due care" principle is being violated. A professional accountant shall comply with the principle of professional competence and due care, which requires an accountant to attain and maintain professional knowledge and skill at the level required to ensure that a client or employing organization receives competent professional service, based on current technical and professional standards and relevant legislation; and act diligently and in accordance with applicable technical and professional standards.

Diligence includes responsibility to act carefully, thoroughly and on a timely basis in accordance with requirements of an assignment.

(iii) In the given situation, CA Jatin's client instructs him to intentionally understate the sales figures to reduce the tax liability. It violates "integrity" principle. A professional accountant shall comply with the principle of integrity, which requires an accountant to be straightforward and honest in all professional and business relationships. Integrity implies fair dealing and truthfulness.

A professional accountant shall not knowingly be associated with reports, returns, communications or other information where the accountant believes that the information contains a materially false or misleading statement; contains statements or information provided negligently or omits or obscures required information where such omission or obscurity would be misleading.

Preference dividend	2,50,000 x 12%	30,000
PAES		41,250
EPS (ratio measures overall profits generated for each share in existence)	41,250 / 1,00,000	0.41
Debt Equity ratio (indicates composition of capital structure in terms of debt and equity)	2,50,000 / 16,25,000	0.15
MPS	FV 10 + 20% Premium	12
Market value to Book value (indicates market response of the equity shareholders' investment)	12 Lakhs / 13.75 Lakhs	0.87
PE Ratio (indicates expectations of the equity investors about the earnings of the firm)	12/0.41	29.27 times

Issue	Ratio	Numerator	Denominator	Ideal Ratio/ Range	Ratio of the Company	Favourable/ Adverse
1	Current Ratio	Current Assets	Current Liabilities	2	1.6	Adverse
2	Quick Ratio	Quick Assets	Current Liabilities	1	0.32	Adverse
3	DSCR	Earnings available for Debt services	Interest & Instalments	2	6.1	Favourable
4	Interest coverage ratio	EBIT	Interest	> 1	4.8	Favourable
5	Preference Dividend coverage ratio	PAT	Preference dividend	> 1	2.375	Favourable
6	Fixed charges coverage ratio	EBIT + Depn	Interest + Repayment of loan	> 1	6.34	Favourable

7	Debt equity ratio	Debt	Equity	Not applicable	0.15	
8	EPS	Profit available for equity holders	No. of Equity Shares	Not applicable	0.41	
9	MV/BV	MV of Equity Shares	BV of Equity Shares	Not applicable	0.87	

Earnings available for Debt services = PAT + Interest + Depreciation + Instalments

= 71,250 + 37,500 + 3,75,000 + 50,000 = 5,33,750

6. (i) Cost of Equity Share capital

As per CAPM Model K_e = $R_f + ß (R_m - R_f)$

Rf = 6%

B = 1.54

Rm-Rf = 8%

Ke = 6% + 1.54(8%)

Ke = **18.32%**

(ii) Cost of Preference Share capital

n = 8

Net Proceeds (NP) = 150

Redemption Value (RV) = 100

Preference Dividend (PD) = 15

$$K_p = \frac{PD + \frac{(RV - NP)}{n}}{\frac{(RV + NP)}{2}}$$

K_p $= \dfrac{15 + \left(\dfrac{100 - 150}{8}\right)}{\left(\dfrac{100 + 150}{2}\right)}$

K_p = **7%**

(iii) Cost of convertible debenture

Cash Redemption Value (RV)	= 100
Share Redemption Value (RV):	
Value of share after 5 years	= 25 x $(1.06)^5$ = 33.455
Share Redemption Value (RV)	= 33.455 x 4 = 133.82

Therefore, investor will choose share redemption.

Redemption Value (RV)	= 133.82
Net Proceeds (NP)	= 100
n	= 5
Interest (I)	= 13
Tax (t)	= 25%

K_d $= \dfrac{I(1-t) + \dfrac{(RV - NP)}{n}}{\dfrac{(RV + NP)}{2}}$

$= \dfrac{13(1-0.25) + \dfrac{(133.82-100)}{5}}{\dfrac{(133.82+100)}{2}}$

K_d = **14.13%**

(iv) Cost of Retained Earnings

$K_r = K_e$ = **18.32%**

(v) Cost of Term Loan

= 11% x (1-0.25) = **8.25%**

7. Market Value of Equity = ₹ 25,00,000

K_e = 21%

$$\frac{\text{Net income (NI) for equity - holders}}{K_e} = \text{Market Value of Equity}$$

$$\frac{\text{Net income (NI) for equity holders}}{0.21} = 25,00,000$$

Net income for equity holders = 5,25,000

EBIT= 5,25,000/0.7 = 7,50,000

	All Equity	Debt and Equity
EBIT	7,50,000	7,50,000
Interest to debt-holders	-	(75,000)
EBT	7,50,000	6,75,000
Taxes (30%)	(2,25,000)	(2,02,500)
Income available to equity shareholders	5,25,000	4,72,500
Income to debt holders plus income available to shareholders	5,25,000	5,47,500

Present value of tax-shield benefits = ₹ 5,00,000 × 0.30 = ₹ 1,50,000

(i) Value of Restructured firm

= ₹ 25,00,000 + ₹ 1,50,000 = ₹ 26,50,000

(ii) Cost of Equity (K_e)

Total Value = ₹ 26,50,000

Less: Value of Debt = ₹ 5,00,000

Value of Equity = ₹ 21,50,000

$K_e = \frac{4,72,500}{21,50,000} = 0.219 = 21.98\%$

(iii) WACC (on market value weight)

Cost of Debt (after tax) = 15% (1- 0.3) = 0.15 (0.70) = 0.105 = 10.5%

Components of Costs	Amount	Cost of Capital (%)	Weight	WACC (%)
Equity	21,50,000	21.98	0.81	17.80
Debt	5,00,000	10.50	0.19	2.00
	26,50,000			19.80

Comment: At present the company is all equity financed. So, $K_e = K_o$ i.e. 21%. However, after restructuring, the K_o would be reduced to 19.80% and K_e would increase from 21% to 21.98%.

8. **(i) Degree of operating, financial and combined leverages at the current sales level-**

$$DOL = \frac{\text{Contribution}}{\text{EBIT}}$$

$$= \frac{₹\,34,00,000 - ₹\,6,00,000}{₹\,22,00,000} = 1.27$$

$$DFL = \frac{\text{EBIT}}{\text{EBT}}$$

$$= \frac{₹\,22,00,000}{₹\,16,00,000} = 1.375$$

DCL = DOL × DFL = 1.27 × 1.38 = 1.75

(ii) Earnings per share at the new sales level **(Amount in ₹)**

Particulars	Increase by 20%	Decrease by 20%
Sales level	40,80,000	27,20,000
Less: Variable expenses	7,20,000	4,80,000
Less: Fixed cost	6,00,000	6,00,000

Earnings before interest and taxes	27,60,000	16,40,000
Less: Interest	6,00,000	6,00,000
Earnings before taxes	21,60,000	10,40,000
Less: Taxes @35%	7,56,000	3,64,000
Earnings after taxes (EAT)	14,04,000	6,76,000
Number of equity shares	8,00,000	8,00,000
EPS	1.76	0.85

Working Notes:

Variable Costs = ₹ 6,00,000 (total cost - depreciation)

Variable Costs at:

(i) Sales level, ₹ 40,80,000 = ₹ 7,20,000 (increase by 20%)

(ii) Sales level, ₹ 27,20,000 = ₹ 4,80,000 (decrease by 20%)

9. Under MIRR, if the cash flows are equal over the life of the asset

Initial investment x $(1 + \text{MIRR})^n$ = CFAT x FVAF

Terminal Value (CFAT x FVAF) = ₹ 11,43,589

Initial investment x $(1+10.9\%)^8$ = ₹ 11,43,589

Initial investment = 11,43,589 / $(1+10.9\%)^8$ = 5,00,000 (rounded off to nearest lakh)

Under simple payback period, if the cash flows are equal over the life of the asset

$$\text{Payback Period} = \frac{\text{Total initial capital investment}}{\text{Annual expected after - tax net cash flow}}$$

Initial investment = CFAT x Simple payback period in years

5,00,000 = 5 x CFAT

CFAT = 1,00,000

Under Discounted payback period, if the cash flows are equal over the life of the asset

PVAF $= \dfrac{\text{Total initial capital investment}}{\text{Annual expected after - tax net cash flow}}$

Initial investment = CFAT x PVAF @ WACC% for 7 years

5,00,000 = 1,00,000 x PVAF @ WACC% for 7 years

PVAF @ WACC% for 7 years = 5

So, WACC = 9%

NPV = Sum of Present Values of Cash inflows – Cost of the Project

= 1,00,000 x 5.535 – 5,00,000 = 53,500

Profitability Index $= \dfrac{\text{Sum of present value of net cash inflow}}{\text{Initial cash outflow}}$

$= \dfrac{₹5,53,500}{₹\ 5,00,000} = 1.107$

PAT = CFAT – Depreciation

$= 1,00,000 - \dfrac{5,00,000}{8} = 37,500$

ARR $= \dfrac{\text{Profit after Depreciation}}{\text{Investment in the begining of the year}} \times 100$

$= \dfrac{37,500}{5,00,000} = 7.5\%$

Appraisal technic	**Acceptable range**	**Outcome for this asset**	**Recommendation**
NPV	> 0	53,500	Accept
MIRR	MIRR > WACC	10.9% > 9%	Accept
Simple payback	Payback < Life	5 < 8	Accept
Discount payback	Payback < Life	7 < 8	Accept
ARR	ARR > WACC	7.5% < 9%	Reject
PI	> 1	1.107	Accept

10. 1. Price under Walter's Model (P) $= \dfrac{D + \dfrac{r}{K_e}(E - D)}{K_e}$

	Situation 1	Situation 2
1. DP Ratio	40%	60%
2. Retention Ratio	60%	60% - 20% = 40%
3. Dividend per share = D	₹ 4	₹ 6
4. Retention per share = (E-D)	₹ 6	₹ 4
5. Market price per share = MPS	₹ 57.50	₹ 57.50 – ₹ 2.50 = ₹ 55
6. Equation = Let r/ke be y & Ke be x.	4 + 6y = 57.50x	6 + 4y = 55x
7. Equation for solving	57.50x – 6y = 4	55x – 4y = 6

Equation 1 * 4 = 230x – 24y = 16 ....(1)

Equation 2 * 6 = 330x – 24y = 36 ....(2)

By solving, we get x = 20% and y = 1.25.

$y = \dfrac{r}{K_e} = 1.25$

$\dfrac{r}{20\%} = 1.25.$

Hence, r = 25%

2. Since r 25% > ke 20%, the company should retain 100%. Since the company pays 40% Dividend, the company is not following optimum dividend policy.

3.

	Situation 1	Situation 2
1. DP Ratio	0%	100%
2. Retention Ratio	100%	0%
3. Dividend per share = D	₹ 0	₹ 10
4. Retention per share = (E-D)	₹ 10	₹ 0

5. Market price per share = MPS	$\frac{10 \times 1.25}{0.2} = 62.50$	$\frac{10}{0.2} = 50$

4. Price under Gordon's Model

$$\text{Market price per share}(P_0) = \left[\frac{D_0(1+g)}{K_e - g}\right]$$

	Situation 1	Situation 2
1. Dividend paid	₹ 4	₹ 6
2. Retention Ratio (b)	60%	40%
3. Return on investment (r)	25%	25%
4. Growth = g = br	15%	10%
5. Market price per share = MPS	$\frac{4(1+0.15)}{0.2-0.15}$ = ₹ 92	$\frac{6(1+0.10)}{0.2-0.1}$ = ₹ 66

5. Since, return on investment is used for computing growth rate, both cost of equity and return on investment are considered for computing price under Gordons model. Hence, the opinion of the chief financial officer is not valid.

11. (a) Features of Bulldog Bond

- It is denominated in Bulldog Pound Sterling/Great Britain Pound (GBP)
- Issued in London
- Issuer Non- UK Company
- Regulations: Great Britain
- Purpose: Access of capital available in UK market
- Issue proceeds can be used to fund UK operation
- Issue proceeds can be used to fund a company's local opportunities

(b) Spontaneous sources of finance are those which naturally arise in the course of business operations. Trade credit, credit from employees, credit from suppliers of services, etc. are some of the examples which may be quoted in this respect.

Spontaneous Sources of Finance

(i) **Trade Credit:** Trade credit is a spontaneous source of finance which is normally extended to the purchaser organization by the sellers or services providers. It contributes to about one-third of the total short-term requirements.

(ii) **Bills Payable:** In the case of "Bills Payable" the purchaser will have to give a written promise to pay the amount of the bill/invoice either on demand or at a fixed future date to the seller or the bearer of the note.

(iii) **Accrued Expenses:** The accrued expenses refer to the services availed by the firm, but the payment for which has yet to be made. It is a built in and an automatic source of finance as most of the services like wages, salaries, taxes, duties etc., are paid at the end of the period.

(c) **Over-capitalisation arises due to following reasons:**

(i) Raising more money through issue of shares or debentures than company can employ profitably.

(ii) Borrowing huge amount at higher rate than rate at which company can earn.

(iii) Excessive payment for the acquisition of fictitious assets such as goodwill etc.

(iv) Improper provision for depreciation, replacement of assets and distribution of dividends at a higher rate.

(v) Wrong estimation of earnings and capitalization

(d) **Disadvantages of Profit Maximisation objective of financial management.**

(i) Emphasizes the short-term gains

(ii) Ignores risk or uncertainty

(iii) Ignores the timing of returns

(iv) Requires immediate resources.

6B: STRATEGIC MANAGEMENT

QUESTIONS

Multiple Choice Questions

1. "The world needs to move fast, at a speed of innovation that makes 6 months look like 60 years" - Jayanta R., the founder of Hepha.io, while speaking at a technology vision conference at Hyderabad.

   In 2009, Jayanta was working as a machine learning research scholar at a US college, when he struck upon an idea of technological research being automated. He thought - what could happen if computers could innovate themselves. The world will change forever. His idea took shape of a boutique AI consultancy in 2020, Hepha.io.

   Hepha.io had one vision, to speed up innovation. And for this they hired the most affluent research scholars from across the globe - all virtually, all paid in equity, with no real organizational structure. The offering to customers was simple - it was futuristic, high value added to existing operations and could be applied to a lot of other processes.

   The company was doing great in terms of customer acquisition and profitability, about 150 of Fortune 500 companies had relations with Hepha.io, bringing in about $12 million in earnings. But the company was disappointed by their own speed of growth, which was contrary to their vision and thereby a joke at many conferences that hurt Jayanta immensely. Hepha.io had by now been seen in the west, as an aggressive, crazy company that wanted to change the world with computers, without caring much about the harms of massive heat generating servers being built on agricultural land, damaging crops, weather and even harming social setups by shattering employment. To move away from west and speed things up internally, they moved their focus to India, the land of technically skilled labor force with ease of doing business.

Entering India helped them dodge environmental concerns, but future scare of unemployment was a great threat. Jayanta met many political influencers and industrialists but could not attract much attention. Until Harsh G., the young scientist of LifeMind Technologies, met him in Mumbai, and offered to help.

LifeMind Tech, had a long running reputation of being a socially awake enterprise, which reinvested its earnings in developing India. Jayanta could not have asked for a better partner in India. This partnership helped Hepha.io to overcome a lot of political backlash and even the sharpest minds in the Indian tech space seemed to join the company without any employment apprehensions. What also helped Jayanta realize under Harsh's guidance was that innovation is necessary to enter the future but not at the cost of destroying the present. Hepha.io has since then opted for India as its Asia headquarters, driven by the purpose of holistic innovation.

Based on the above Case Scenario, answer the Multiple Choice Questions.

(i) After the move to India, Hepha.io decided to align innovation with sustainability under the theme of "Holistic Innovation." Which level of strategy does this shift primarily represent?

- (a) Corporate level strategy
- (b) Business level strategy
- (c) Functional level strategy
- (d) Operational level strategy

(ii) The strategic move by Hepha.io to shift its existing AI services into the new Indian geographical market is a clear application of which Grand Strategy?

- (a) Market Penetration
- (b) Market Development
- (c) Product Development
- (d) Retrenchment

(iii) Hepha.io's initial structure, characterized by a highly dispersed team of scholars, relies heavily on coordination through technology and contracts. This structural design is most accurately identified as:

(a) Divisional Structure

(b) Matrix Structure

(c) Network Structure

(d) Strategic Business Unit (SBU) Structure

(iv) The public perception and ethical backlash concerning environmental damage and technological unemployment, which severely impacted Hepha.io's operations in the West, are analysed under which factor of the PESTLE analysis?

(a) Political

(b) Technological

(c) Economic

(d) Socio-Cultural

(v) Hepha.io's founding premise of offering "futuristic, high value add" AI solutions as its primary client appeal is an example of establishing a Competitive Advantage through:

(a) Cost Leadership, by optimising production costs.

(b) Differentiation, by providing a uniquely valuable product.

(c) Focused Cost Leadership, by targeting a narrow segment with the lowest price.

(d) Core Values, by adhering to ethical standards.

2. A mid-sized electronics firm notices a rapid increase in demand for smart home devices. It has decided to redesign its product line to cater specifically to this trend while competitors are still hesitant. Which strategic approach is the firm using?

(a) Reactive strategy

(b) Proactive strategy

(c) Retrenchment strategy

(d) Stability strategy

3. EcoRide Motors, an Indian electric scooter company, faces rising competition from new EV start-ups offering affordable models. Battery suppliers have raised costs due to global shortages, while customers now compare multiple brands online before purchase. Traditional petrol scooter manufacturers are also entering the EV segment. EcoRide struggles to maintain profitability despite growing market demand. Which force poses the greatest threat to EcoRide's profitability?

   (a) Threat of Substitutes

   (b) Bargaining Power of Suppliers

   (c) Rivalry among Existing Competitors

   (d) Bargaining Power of Buyers

4. A financial services company wants to launch a mobile-first platform. The strategy is clearly defined and systems for operations are automated. However, employees are unfamiliar with the technology, and the leadership style emphasizes strict hierarchy, slowing down decision-making. Management wants to ensure the new digital initiative succeeds without resistance or delays. Which 7S elements need urgent alignment?

   (a) Skills and Style

   (b) Structure and Staff

   (c) Shared Values and Systems

   (d) Strategy and Structure

5. In the Indian automobile industry, firms like Speedo Motors and Turbo Auto compete in the mid-range car segment, focusing on fuel efficiency. Meanwhile, LuxeDrive and Elite Motors dominate the premium segment, emphasizing luxury and advanced technology. Each group follows distinct pricing and marketing strategies and movement between these segments is limited due to brand perception and R&D costs. What does this scenario best illustrate?

   (a) Strategic Alliances

(b) Strategic Group Mapping

(c) Value Chain Analysis

(d) Core Competence Identification

6. TechNova Inc., a leading U.S. robotics company, partners with Mechtron India Pvt. Ltd. to jointly design and manufacture affordable industrial robots for the South Asian market. Both companies agree to share technology, production facilities and marketing channels to accelerate market entry and reduce costs. What type of strategy is TechNova implementing?

   (a) Joint Venture

   (b) Market Penetration

   (c) Strategic Alliance

   (d) Diversification

Descriptive Questions

Chapter 1-Introduction to Strategic Management

7. **AquaPure Beverages** is a reputed bottled water and healthy drinks company facing fierce competition from new entrants offering organic, flavoured and eco-friendly alternatives. This has made it challenging to retain customers as established soft drink brands are also launching their own health-focused beverage lines. The company has appointed Ms. Somya to lead the company forward as the Sales & Marketing Manager. Ms. Somya needs to design creative and innovative advertising campaigns to gain a competitive edge, engage the public and capture the market.

   Identify the strategic level that represents Somya's role at AquaPure Beverages. As a strategic advisor, highlight the various benefits of strategic management in overcoming different challenges to Ms. Somya.

8. 'A company's mission statement is typically focused on its present business scope.' Explain the significance of a mission statement.

Chapter 2-Strategic Analysis: External Environment

9. **Riya Sharma** owns a confectionery business in Jaipur, specializing in homemade chocolates and candies. Despite holding a substantial market share in the central region, her business has experienced declining sales of these products over the last few years. Concerned about the market dynamics, Riya consults a management expert for guidance. The consultant recommends a comprehensive understanding of the competitive landscape. Explain the steps to be followed by Riya Sharma to understand the competitive landscape to address the sales decline.

10. As per Michael Porter's Five Forces Model of Competition, intense rivalry among existing competitors reduces the overall attractiveness and profitability of an industry. In this context, explain the situations in which competition among firms becomes cut-throat, leading to lower industry profitability.

Chapter 3-Strategic Analysis: Internal Environment

11. **EcoBuild Constructions,** a leading infrastructure development company, is planning to launch a large-scale smart city project. The project involves multiple stakeholders, including government bodies, investors, suppliers, local communities and environmental groups — each with varying levels of influence and interest. To ensure smooth project execution and stakeholder cooperation, the management wants to analyze and manage these stakeholders effectively. How can EcoBuild Constructions use Mendelow's Matrix to analyze and manage stakeholders in this project?

12. Write a short note on the key strategic drivers of an organization.

Chapter 4-Strategic Choices

13. **ReInnovate Electronics Ltd.,** a company engaged in the manufacture of consumer electronic appliances since 2010, had been performing well until 2022. However, thereafter its market share began to decline steadily. The company started incurring accumulated losses which adversely affected its cash flow position. Consequently, employee morale also deteriorated. The Board of Directors (BoD) of the company

decided to continue its operations by focusing on improving internal efficiency. To achieve this, the BoD is formulating a practical action plan aimed at bringing about a radical change in strategic direction including a revamp of the top management team. Which strategy should the company adopt in this situation? Also, outline the stages involved in the action plan for implementing this strategy.

14. Explain the 'product market growth matrix' as propagated by Igor Ansoff as a device for identifying growth opportunities for the future.

Chapter 5-Strategy Implementation and Evaluation

15. **BrightWave Solutions**, a small and medium-sized company in the renewable energy sector, wants to adopt the latest digital technologies to improve its operational processes and product offerings. The company aims to gain a competitive edge in the rapidly evolving renewable energy market through digital transformation. BrightWave Solutions also seeks to manage continuous changes effectively while transforming its management practices and workflows. Identify the strategy required for successful digital transformation. Also, state the most preferred practices that the company should follow.

16. Explain how organizations can effectively manage strategic uncertainties in a rapidly changing business environment.

SUGGESTED ANSWERS/HINTS

1. **(i)** **(a)** **(ii)** **(b)** **(iii)** **(c)** **(iv)** **(d)** **(v)** **(b)**

2. **(b)**

3. **(c)**

4. **(a)**

5. **(b)**

6. **(c)**

7. Somya's role at AquaPure Beverages represents the **Functional level of strategy.** As the sales and marketing manager, her responsibilities are focused on specific areas within the company, particularly on crafting and executing marketing and sales strategies that drive customer engagement and competitive positioning.

Benefits of Strategic Management for Ms. Somya at AquaPure Beverages

Strategic management can provide several benefits to Somya in addressing the competitive and consumer challenges faced by AquaPure Beverages:

- Strategic management helps AquaPure Beverages define its goals and mission, providing clear **direction for future initiatives.** This ensures that all marketing efforts are aligned with the company's overall vision. It allows Somya to **set realistic** and achievable **objectives** that support the company's **long-term goals** ensuring that marketing strategies are both ambitious and attainable.
- Through strategic management, Somya can **proactively shape the future** of AquaPure Beverages rather than merely reacting to market changes. This allows the company to **anticipate trends and act accordingly.** A proactive approach enables AquaPure Beverages to better manage environmental uncertainties and stay ahead of competitors, ensuring a more controlled and predictable business environment.
- Strategic management provides a robust **framework for making critical decisions** regarding marketing strategies, target markets and resource allocation. This ensures that all major decisions are well-informed and strategically sound. It ensures **coherence and consistency in decision-making** across the organization, aligning marketing strategies with overall business objectives.
- **Strategic management helps identify and exploit new business opportunities,** allowing Somya to craft campaigns that resonate with emerging consumer preferences and market trends. By recognizing and capitalizing on these opportunities, AquaPure

Beverages can differentiate itself from competitors and capture a larger market share.

- **Strategic management acts as a defence mechanism against potential mistakes and pitfalls,** helping Somya avoid costly errors in marketing decisions and campaign execution. It provides a structured approach to identifying and mitigating risks, ensuring more informed and safer decision-making.
- **Strategic management enhances the longevity and sustainability of AquaPure Beverages** by ensuring that marketing strategies are adaptable and resilient in a dynamic market. It helps the company establish a clear and deliberate position within the industry, ensuring sustained relevance and competitiveness.
- **Strategic management enables the development of core competencies and competitive advantages** that are crucial for the company's success. This includes building strong brand identity, innovative product offerings, and superior customer service. By focusing on these strengths, Somya can ensure that AquaPure Beverages achieves sustainable growth and maintains its competitive edge in the beverage industry.

Through strategic management, Somya can effectively navigate the competitive challenges faced by AquaPure Beverages. By providing clear direction, encouraging a proactive approach, guiding critical decisions, identifying new opportunities, defending against pitfalls, ensuring longevity and developing core competencies, strategic management enables the company to achieve and sustain a competitive edge. This comprehensive approach will allow Somya to design innovative advertising campaigns that engage the public, capture the market, and drive the company forward.

8. A company's mission statement is typically focused on its present business scope, **who we are and what we do**. Mission statements broadly describe an organization's present capability, customer focus, activities, and business make up. Mission for an organization is significant for the following reasons:

- It ensures **unanimity of purpose** within the organization.
- It develops a basis, or standard, for **allocating organizational resources.**
- It provides a basis for **innovating the use of the organisation's resources**
- It **establishes** a general tone or **organizational climate**, to suggest a business like operation.
- It serves as a **focal point** for those who can identify with the **organisation's purpose and direction**.
- It facilitates the **translation of objectives and goals into a work structure** involving the assignment of tasks to responsible elements within the organization.
- It specifies organizational purposes and the **translation of these purposes into goals** in such a way that cost, time, and performance parameters can be assessed and controlled.

9. Steps to understand the competitive landscape are as follows:

(i) **Identify the competitor:** The first step to understanding the competitive landscape is to identify the competitors in the firm's industry and have actual data about their respective market share.

(ii) **Understand the competitors:** Once the competitors have been identified, the strategist can use market research report, internet, newspapers, social media, industry reports, and various other sources to understand the products and services offered by them in different markets.

(iii) **Determine the strengths of the competitors:** What is the strength of the competitors? What do they do well? Do they offer great products? Do they utilize marketing in a way that comparatively reaches out to more consumers. Why do customers give them their business?

(iv) **Determine the weaknesses of the competitors:** Weaknesses (and strengths) can be identified by going through consumer reports and reviews appearing in various media. After all, consumers are often willing to give their opinions, especially when the products or services are either great or very poor.

(v) **Put all of the information together:** At this stage, the strategist should put together all information about competitors and draw inference about what they are not offering and what the firm can do to fill in the gaps. The strategist can also know the areas which need to be strengthened by the firm.

10. According to Michael Porter's Five Forces framework, rivalry among competitors significantly influences the attractiveness and profitability of an industry. When rivalry becomes cutthroat, several conditions contribute to low industry profitability:

1. **Industry Leader Presence**: While a strong industry leader can help maintain pricing discipline, the effectiveness diminishes as the number of competitors increases. Many rivals can engage in aggressive pricing strategies, leading to decreased profitability.

2. **Number of Competitors**: A higher number of competitors increases rivalry, making it difficult for any single firm to control pricing. This leads to intensified price competition, which adversely affects industry profitability.

3. **High Fixed Costs**: Industries with high fixed costs create pressure on firms to fully utilize their capacity. When firms face excess capacity, they often resort to price cuts to maintain sales volume, which diminishes profitability across the industry.

4. **Exit Barriers**: High exit barriers prevent firms from leaving the industry, keeping competition high. Specialized assets or other constraints can lead firms to remain in the market, maintaining competitive pressure and negatively impacting profitability for all players.

5. **Product Differentiation**: In industries lacking product differentiation, firms primarily compete on price. This leads to price wars and lower profitability. In contrast, firms that can

differentiate their products tend to achieve higher profit margins and reduce competitive pressure.

6. **Slow Industry Growth**: When industry growth slows, firms may adopt aggressive tactics to protect or gain market share, resulting in intensified rivalry and diminished profitability as they compete for a limited customer base.

In summary, conditions such as the presence of a strong industry leader, the number of competitors, high fixed costs, exit barriers, lack of product differentiation and slow industry growth contribute to cutthroat rivalry and low industry profitability.

11. EcoBuild Constructions can use **Mendelow's Matrix** effectively to analyze and manage stakeholders through a structured, grid-based approach by following these steps:

1. **Identify Stakeholders:** Begin by identifying all relevant stakeholders such as government bodies, investors, suppliers, environmental groups and local communities who may influence or be affected by the project.

2. **Assess Power and Interest:** Evaluate each stakeholder's **power** (ability to influence project outcomes) and **interest** (level of concern or involvement). Power may stem from authority, resources or expertise, while interest relates to the stakeholder's stake in the project's success.

3. **Plot Stakeholders on the Grid:** Create a grid with Power on one axis and Interest on the other. Plot stakeholders accordingly:
   - **High Power–High Interest:** Key Players
   - **High Power–Low Interest:** Keep Satisfied
   - **Low Power–High Interest:** Keep Informed
   - **Low Power–Low Interest:** Low Priority

4. **Develop Strategies for Each Quadrant:**
   - **Key Players:** Engage closely with these stakeholders, involve them in decision-making and keep them informed as their support is vital for project success.

- **Keep Satisfied:** Provide periodic updates and address concerns to maintain their satisfaction and prevent potential opposition.
- **Keep Informed:** Communicate regularly and share progress updates to sustain their interest and gain valuable feedback.
- **Low Priority:** Monitor occasionally for changes in their power or interest but allocate minimal resources to manage them.

5. **Monitor and Adapt:** Continuously review stakeholder power and interest, as these may change over time. Adjust engagement strategies accordingly to ensure ongoing alignment with project goals.

By applying **Mendelow's Matrix**, EcoBuild Constructions can systematically analyze and manage stakeholders, tailor communication and engagement efforts and build strong relationships—thereby increasing the likelihood of the project's success.

12. Key Strategic Drivers of an Organization

Strategic drivers are essential elements that influence an organization's ability to differentiate itself from its competitors and achieve competitive advantage. These drivers assess the current performance of the business and provide insights into areas that need focus.

The key strategic drivers include:

1. **Industry and Markets:** Understanding the industry and markets is crucial for identifying the organization's relative position. Industries group similar companies based on their primary products, while markets are defined by the buyers and sellers of these products. Analyzing industry and market dynamics, often through tools like strategic group mapping, helps organizations evaluate competition and refine strategies.
2. **Customers:** Identifying and understanding customers is a critical driver. Customers are segmented based on their needs and spending capacity, which guides product development and marketing strategies. Differentiating between customers (buyers)

and consumers (users) is vital to tailoring pricing, design, and usability strategies effectively.

3. **Products and Services:** Products and services are central to defining the business. Organizations must assess their offerings, classify products, and devise strategies for differentiation, branding, and pricing. Product innovation and marketing are key to maintaining competitiveness.

4. **Channels:** The channels through which products and services are delivered impact accessibility and customer satisfaction. Strategies related to direct, digital, or relationship-based marketing ensure the efficient distribution of offerings to target customers.

By aligning these drivers with organizational goals, businesses can achieve sustained growth and maintain a competitive edge.

13. In the given situation, the suitable **retrenchment strategy** for **ReInnovate Electronics Ltd.** is the **Turnaround Strategy**. This strategy is designed to reverse the company's decline and restore it to profitability, particularly when the company is facing challenges such as declining market share, negative cash flows, and low employee morale.

Stages in the Action Plan for Turnaround Strategy:

- **Stage One - Assessment of Current Problems**: The first step is to assess and diagnose the root causes of the company's decline, such as uncompetitive products, poor cash flow or internal inefficiencies. This stage involves determining the extent of the damage caused by these problems.

- **Stage Two - Analyze the Situation and Develop a Strategic Plan**: Evaluate the chances of the company's survival and develop a strategic plan that outlines the corrective actions to be taken. This includes identifying appropriate strategies to address internal inefficiencies, improving competitiveness and enhancing employee morale.

- **Stage Three - Implementing an Emergency Action Plan**: If the situation is critical, an immediate action plan must be implemented to stabilize the business. This may involve cutting costs, ensuring positive cash flow, raising necessary funds and addressing short-term operational issues.

- **Stage Four - Restructuring the Business**: Focus on restructuring the company's core business operations, especially if they have been significantly affected. This stage involves efforts to improve efficiency, restructure finances and position the company for long-term recovery and growth.
- **Stage Five - Returning to Normal**: In the final stage, the company should begin showing signs of profitability and improving financial performance. Strategic efforts such as introducing new products, improving customer service and forming alliances should be emphasized to restore market share and build long-term sustainability.

By following these stages, **ReInnovate Electronics Ltd.** can develop a comprehensive turnaround plan to regain financial stability, improve operational efficiency and rebuild its competitive position in the market.

14. Ansoff's Product Market Growth Matrix, developed by Igor Ansoff, is a strategic tool that helps businesses identify growth opportunities by analyzing the interplay between products and markets.

It offers four distinct strategies based on whether the products and markets are existing or new. These strategies are:

1. **Market Penetration**: Focuses on selling existing products in existing markets. This involves increasing market share by enhancing sales through advertising, promotions, competitive pricing, or encouraging higher usage among current customers.
2. **Market Development**: Entails selling existing products in new markets. This could involve exploring new geographical regions, utilizing alternative distribution channels, or creating new market segments.
3. **Product Development**: Involves introducing new or modified products into existing markets. This strategy often requires innovation and developing products that meet current market needs.

4. **Diversification**: Refers to marketing new products in new markets. It is a high-risk strategy as the business ventures into unfamiliar products and markets.

As market dynamics evolve, companies may transition between these strategies to adapt and sustain growth. The matrix provides a structured framework for businesses to align their growth strategies with their capabilities and market conditions.

15. To enable successful digital transformation, the appropriate strategy for **BrightWave Solutions** is a **Change Management Strategy**. This strategy ensures that the organizational, procedural and cultural changes associated with adopting new digital technologies are effectively managed.

Following are the five best practices for managing change in small and medium-sized businesses:

1. **Begin at the top:** Change should be initiated and driven by unified, focused leadership. A committed top management can create and promote a culture that inspires the organization to embrace change.

2. **Ensure that the change is both necessary and desired:** Before initiating digital transformation, decision-makers must understand its necessity and long-term impact. Without a proper strategy, introducing too much change too quickly can be counterproductive.

3. **Reduce disruption:** It is essential to minimize employee disruption during transformation. This can be done by:

- Communicating changes early.
- Equipping employees with tools and training.
- Empowering change agents like project managers.
- Involving IT teams proactively.
- Creating an environment conducive to change.

4. **Encourage communication:** Promote open and continuous two-way communication. This ensures that employees feel heard and involved and that concerns are addressed timely. Effective communication also promotes innovation and collaboration across departments.

5. **Recognize that change is the norm, not the exception:** Businesses must treat change as a continuous process, not a one-time project. Being change ready helps the organization to adapt, sustain performance and stay aligned with evolving customer expectations.

By following these practices, **BrightWave Solutions** can successfully manage its digital transformation, improve operational efficiency, enhance product offerings and maintain a competitive edge in the renewable energy industry.

16. In managing strategic uncertainties in a rapidly changing business environment, organizations need to adopt proactive strategies to navigate unpredictability effectively. Following are the several key approaches:

Flexibility: Organizations should build flexibility into their strategies to enable quick adaptation to change in the environment.

Diversification: Diversifying the organization's product portfolio, markets, and customer base can help reduce the impact of strategic uncertainty.

Monitoring and Scenario Planning: Regularly monitoring key indicators of change and conducting scenario planning exercises can help organizations anticipate and prepare for different future scenarios.

Building Resilience: Investing in building internal resilience is essential for weathering uncertainty. This includes strengthening operational processes, increasing financial flexibility, and improving risk management capabilities.

Collaboration and Partnerships: Collaborating with other organizations, suppliers, customers, and partners can provide access to additional resources, expertise, and market opportunities. Strategic partnerships enable organizations to pool resources, share risk, and leverage each other's strengths to navigate uncertainty more effectively.